



AI-Related Human Existential Risk and the Civilizational Subject

From Species Survival to Civilizational Subjecthood

Species, Civilization, and Philosophy in the Age of AGI

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Abstract

This paper develops an ontological and conceptual-engineering framework for understanding AI-related human existential risk. As AI systems become more capable, more general-purpose, more agentic, and more deeply embedded in human knowledge production, institutional processes, judgment formation, and technological development, AI risk can no longer be understood merely as a problem of model performance, product safety, or engineering governance. In frontier technical contexts, AI is increasingly used to assist code generation, model evaluation, research automation, and the development of AI systems themselves. In public-ethical contexts, AI is also increasingly understood as posing categorical challenges to human dignity, responsibility, freedom, the common good, and the self-understanding of human societies. These developments reopen the question of human existential risk. Yet in many discussions of AI risk, a more basic question remains insufficiently clarified: when we say that AI may pose an existential risk to humanity, what exactly is the “humanity” that is at risk?

This paper argues that AI-related human existential risk should not be assessed solely in terms of whether *Homo sapiens*, as a biological species, is physically destroyed, permanently disempowered, or deprived of its future. Human beings certainly exist as a biological species, but humanity also exists as a civilizational

subject: an intergenerational common world constituted by language, knowledge, institutions, responsibility, truth, agency, value, and meaning. If AI does not directly destroy the human species, but irreversibly weakens the basic conditions under which humanity can continue to exist as a civilizational subject, then this form of risk should also be included within an expanded understanding of AI-related human existential risk.

To develop this claim, this paper distinguishes three nested layers: the human species as the biological substrate of civilization; human civilization as the intergenerational common operating environment formed by the species; and philosophy as the reflective kernel through which civilization defines and revises its foundational categories. The human species provides the biological substrate of civilization. Human civilization enables the species to become a collective subject capable of knowledge, judgment, responsibility, transmission, and meaning-making. The philosophical kernel supplies foundational categories—such as truth, evidence, responsibility, agency, value, and meaning—that are presupposed by civilizational systems such as science, law, politics, education, economy, culture, and the arts.

This paper uses a functional hardware–kernel–distribution analogy to clarify the relation among these layers. The human species may be compared to hardware; philosophy to a kernel; and human civilization to a distribution or complete operating environment running on a biological substrate and structured by a philosophical kernel. The analogy is neither literal nor historical. Its purpose is to illuminate the layered structure of human existence: hardware may remain intact while the system fails to operate well; a distribution may still boot while the kernel is damaged; the application layer may continue to flourish while foundational categories deteriorate.

It concludes by proposing an expanded definition: AI-related human existential risk arises not only when AI threatens the biological survival of humanity, but also when AI irreversibly weakens the civilizational conditions under which humanity can continue to exist as a subject of truth, responsibility, agency, and meaning. This definition does not attempt to classify all AI-driven social change as existential risk. Rather, it provides a conceptual basis for further analysis of the major dimensions of human existential risk in the age of AI and AGI.

Keywords

artificial intelligence; AGI; human existential risk; AI-related human existential risk; civilizational subject; civilizational subjecthood; human species; human civilization; philosophical kernel; conceptual engineering; philosophy of AI; philosophy of technology; civilizational risk

1 | Introduction: AI Reopens the Question of the Subject of Human Existential Risk

Artificial intelligence is reopening the question of human existential risk. By “reopening,” I do not mean merely that AI systems are becoming more capable. Human history has seen many technologies capable of profoundly reshaping civilizational structures, including writing, printing, steam power, electricity, nuclear energy, the internet, and biotechnology. Powerful technologies do not automatically constitute existential risks. What makes contemporary AI distinctive is that its expanding capabilities are entering the very structures through which human beings know the world, form judgments, organize action, allocate responsibility,

govern institutions, and understand themselves. AI is no longer merely a tool for extending external capacities; it is becoming a mediator of human knowledge, institutions, action, and meaning.

Current AI systems are no longer limited to producing textual answers to isolated questions. They can generate code, organize evidence, draft reports, assist research, structure workflows, recommend decisions, call tools, coordinate tasks, and participate in institutional processes across business, education, scientific research, law, medicine, administration, and public services. In frontier research contexts, AI is also increasingly involved in the development of AI systems themselves, including assisted coding, model evaluation, research automation, and technical iteration.[Note 1] This does not mean that AGI, in any strong sense, has already been achieved. Nor does it mean that any widely accepted threshold for AGI has been crossed. The argument of this paper does not depend on such a claim. By “the age of AGI,” I refer more precisely to an unfolding historical transition: AI systems are becoming more general-purpose, more autonomous, more agentic, more deeply embedded in institutions, and increasingly involved in mediating human knowledge, judgment, action, and meaning.

For this reason, the risks discussed in this paper are situated against the background of the age of AGI, but they do not depend on the premise that “true AGI” has already arrived. Many risk mechanisms do not need to wait until some generally accepted definition of AGI has been satisfied. Once AI systems are sufficiently capable, sufficiently widespread, and sufficiently embedded in the structures of human knowledge, institutions, judgment, and action, they have already begun to alter the conditions under which human beings exist as a civilizational subject. More capable, more general, and more autonomous future systems may amplify, generalize, or even render these risks irreversible, but early mechanisms can already be observed in current advanced AI, generative AI, agentic AI, and institutionalized AI applications.

This is why AI risk today is not only a technical problem, but also a philosophical one. Technical questions usually concern whether systems are reliable, safe, controllable, and compliant. Philosophical questions go further: What is truth? Who is a subject? What is responsibility? What has value? Why should human beings still judge and act for themselves? When AI becomes an interface to knowledge, it affects not only what human beings know, but also how they know. When AI becomes an action system, it affects not only whether tasks are completed, but also how actions are formed, attributed, and audited. When AI enters institutional processes, it affects not only efficiency, but also responsibility, trust, and governance. When AI participates in human judgment, it affects not only the outcome of choices, but also whether human beings retain the capacity to form judgment. When AI can generate art, argumentation, research, design, companionship, and strategic advice, it affects not only labor markets, but also how human contribution, creativity, and meaning are understood.

These developments have already pushed AI risk beyond the scope of engineering performance or product safety alone. AI is entering public-ethical domains concerning human dignity, responsibility, freedom, truth, and the common good.[Note 2] Whether or not one accepts the specific premises of any particular religious, philosophical, or ethical tradition, this public discussion itself indicates that AI is no longer merely a problem of tool governance. It has become a question of how human beings can continue to be human.

Yet many discussions of AI existential risk still assume a relatively narrow object of risk. On this view, human existential risk primarily concerns whether AI might, at some future moment, destroy the human species, seize control of critical infrastructure, permanently deprive humanity of its future, or in some decisive sense cause humanity to lose control. These risks are, of course, serious. Human extinction, permanent disempowerment, irreversible catastrophe, and decisive loss of control are among the most direct, basic, and non-negligible forms of existential risk (Bostrom, 2014; Russell, 2019). This paper does not attempt to

diminish their importance.

What this paper argues is that the object of AI-related human existential risk may extend beyond this. The question that must first be clarified is: when we say that AI may threaten “humanity,” what does “humanity” mean? Does it refer only to the continued existence of *Homo sapiens* as a biological species? Does it refer only to human bodies, reproductive capacity, and biological future? Or does humanity also exist as a civilizational subject: as an intergenerational common world constituted by language, knowledge, institutions, responsibility, truth, agency, value, and meaning?

This distinction is not an abstract semantic issue. It is an ontological issue that AI risk theory must address. If humanity is understood only as a biological species, then AI-related human existential risk is largely restricted to physical destruction, species extinction, permanent disempowerment, or loss of control over the future. But if humanity also exists as a civilizational subject, then AI may generate another form of existential harm: the human species may continue to exist, institutions and technical systems may continue to operate, society may continue to produce, consume, and communicate, yet human beings may gradually lose the capacity to know truth, bear responsibility, form judgment, sustain institutions, and create meaning. In such a case, humanity would not be biologically extinct, but it may nevertheless be deeply weakened as a civilizational subject.

This possibility requires us to redefine the bearer of risk before analyzing AI risk. The first step of rational analysis should not be to declare immediately whether AI will or will not destroy humanity. It should be to clarify the structure of the “humanity” that is at stake. If the subject of risk remains unclear, risk classification loses its foundation. Some risks will be mistaken for problems of information quality; others for regulatory delay; others for ordinary tool dependence; still others for cultural anxiety or employment insecurity. Only once we recognize that humanity is not only a species but also a civilizational subject can we ask whether these risks have existential significance.

To this end, the paper distinguishes three nested layers. The first is the human species: *Homo sapiens* as an embodied biological population. This layer includes bodies, genes, nervous systems, perceptual capacities, reproductive mechanisms, physiological vulnerability, and material dependence. The human species is the biological substrate of human civilization. Without this substrate, human civilization cannot continue. The second layer is human civilization: the common operating environment formed by the human species across intergenerational history. It includes language, knowledge, law, institutions, science, education, economy, culture, technology, public reason, responsibility structures, historical memory, and narratives of meaning. Civilization enables the human species not merely to survive, but to judge, cooperate, take responsibility, transmit knowledge, and create meaning. The third layer is the philosophical kernel. Here “philosophy” does not mean only the academic discipline in a narrow institutional sense. It refers to the reflective structure through which civilization defines and revises its foundational categories, including truth, evidence, responsibility, agency, value, and meaning. Civilizational systems such as science, law, politics, education, economy, and culture all have to call upon these foundational categories in order to operate.

This paper uses a functional hardware–kernel–distribution analogy to clarify these three layers. The human species may be compared to hardware; philosophy may be compared to a kernel; and human civilization may be compared to a distribution or complete operating environment running on a biological substrate and structured by a philosophical kernel. The purpose of the analogy is not to technologize human civilization, nor to reduce human beings to machines. It is to clarify the dependency relations among different layers. The existence of hardware does not mean that the system is operating well. A running distribution does not mean that the kernel is stable. A flourishing application layer does not mean that the foundational categories are

healthy. Similarly, the continued existence of human bodies does not mean that humanity remains intact as a civilizational subject; the continued operation of institutions does not mean that responsibility and judgment still genuinely hold; the continued generation of content does not mean that truth remains verifiable.

This three-layer structure also provides ontological coordinates for fragmented discussions of AI risk. Current discussions of AI often slide between different levels: technical research focuses on model alignment, capability control, and system safety; policy discussion focuses on regulation, competition, and institutional adaptation; philosophical discussion focuses on consciousness, agency, and value; social analysis focuses on employment, education, power, and cultural change. These discussions are not unrelated, but they often presuppose different bearers of risk. The distinction among species, civilization, and philosophical kernel forces us first to ask: does a given claim concern the biological survival of the human species, the operating conditions of the civilizational common world, or the stability of civilization's foundational categories? Only within such coordinates can different discussions of AI risk be placed in a common framework that is comparable, connectable, and open to critique.

The argument proceeds as follows. Section 2 examines the limits of understanding human existential risk solely in species-centered terms, arguing that although species extinction is a foundational form of existential risk, it does not exhaust the object of human risk in the age of AI. Section 3 analyzes the human species as the biological substrate of civilization. Section 4 defines human civilization as the intergenerational common world formed by the species, and distinguishes civilization from population, state, culture, and technology. Section 5 argues that philosophy is the categorical kernel through which civilization understands and corrects itself. Section 6 explains the hardware–kernel–distribution analogy and its limits. Section 7 shows why AI makes this three-layer structure a problem for existential risk theory. Section 8 proposes an expanded definition of AI-related human existential risk. Section 9 summarizes the argument and explains the significance of this conceptual reconstruction for AI risk theory, AI governance, and the study of humanity's future.

2 | The Limits of a Species-Centered Understanding of Human Existential Risk

Existing theories of existential risk typically focus on human extinction, permanent disempowerment, the foreclosure of humanity's future, or the loss of humanity's long-term potential (Bostrom, 2013; Ord, 2020). The most direct way to understand human existential risk is therefore to define it as the risk of human species extinction, or as the risk that humanity, as a biological population, permanently loses the capacity to continue into the future. This understanding has an indispensable foundational role. If *Homo sapiens* ceases to exist in the biological sense, or if human beings permanently lose the capacity to reproduce, act, and shape the future, then all further discussion of human civilization, institutions, responsibility, value, and meaning loses its physical bearer. Any theory that seeks to expand the concept of human existential risk must therefore preserve the seriousness of species-level destruction.[Note 3]

Yet necessity is not sufficiency. The continuation of the human species is a necessary condition for the continuation of human civilization, but it does not by itself guarantee that humanity continues to exist as a civilizational subject. Human beings do not exist only as a biological population. Through language, memory, knowledge, institutions, law, ethics, education, art, technology, responsibility, and structures of meaning, human beings also form a common world that extends beyond individual lives and any single generation. If existential risk is restricted to the biological termination of the human species, then many

risks that may leave human beings alive while irreversibly weakening their capacities as a civilizational subject will be systematically excluded from the central discussion.

This exclusion may not be immediately visible in ordinary contexts of technological risk. Under AI conditions, however, it becomes especially important. Current AI systems, and future systems with greater capabilities, need not first threaten human beings by physically destroying them. On the contrary, they are already entering the structures through which human civilization forms knowledge, generates judgment, executes institutions, allocates responsibility, and produces meaning. The resulting question is not only whether AI may kill human beings, but also whether AI may alter the civilizational conditions under which human beings can remain fully human.

2.1 Species Extinction as the Foundational Form of Existential Risk

The destruction of the human species constitutes the foundational form of existential risk because it combines irreversibility, totality, and the foreclosure of the future. Irreversibility means that once the human species becomes extinct, the biological bearer of human civilization no longer exists. Totality means that such a risk is not the failure of a particular region, state, institution, or generation, but the termination of humanity as a biological population as such. The foreclosure of the future means that all future human experience, knowledge, institutions, creation, and value possibilities are closed off.

Species-level existential risk, therefore, is not a category that can be replaced or marginalized. It is the baseline that any broader theory of existential risk must preserve. Without human bodies, nervous systems, reproductive capacities, and intergenerational continuation, there is no continued unfolding of human civilization. Civilization is not an entity floating free of its biological substrate. Any form of human civilization requires some embodied human life as its bearer.

But to treat species extinction as the foundational form of existential risk is not to treat it as the only form. Here it is essential to distinguish necessary conditions from sufficient conditions. The survival of the human species is a necessary condition for the survival of human civilization, but it does not automatically guarantee the continued integrity of human civilization as a common world of subjecthood. Human beings may continue to exist biologically while losing certain basic capacities that constitute such a common world.

This is the basic revision this paper proposes to a purely species-centered understanding of existential risk: if human existential risk is assessed only by asking whether the human species continues to exist, then the biological continuation of human beings will be mistaken for the continuation of humanity as a civilizational subject. In the face of AI as a technology capable of mediating knowledge, judgment, institutions, action, and meaning, this reduction obscures a deeper object of risk.

2.2 Humanity as a Civilizational Subject

Human beings are, of course, a biological species. As *Homo sapiens*, they possess particular bodily structures, nervous systems, perceptual capacities, reproductive mechanisms, and biological vulnerabilities. This level of definition is indispensable from a biological standpoint. Yet if humanity is understood only at this level, we cannot explain why human beings are more than a biological population.

Human beings exist as “human” not only because individuals possess certain physiological structures, but also because, over history, they have formed a common world that exceeds the life span of any individual.

This common world includes language, writing, institutions, law, science, technology, education, ethics, art, religion, philosophy, public reason, and historical memory. It allows individuals not merely to live, but to enter a world already organized, interpreted, and transmitted by previous generations, and within it to learn, judge, commit, create, take responsibility, and ask questions of meaning.

Human civilization is therefore not an external ornament attached to the human species. It is the way in which the human species becomes a civilizational subject. A human individual belongs biologically to the human species, but only through language, relations, norms, education, and shared memory does that individual enter the full human world. Similarly, the human species may continue to survive at a minimal biological level, but if its knowledge systems, institutional systems, responsibility structures, capacities for agency, and structures of meaning are systematically weakened, then humanity's existence as a civilizational subject is also deeply harmed.

This point has direct implications for AI risk theory. AI's threat to humanity need not first appear as an attack on human bodies. It may first appear as a reorganization of civilizational mediating structures: how human beings acquire knowledge, define facts, allocate responsibility, execute institutions, form judgment, and understand labor, creativity, and dignity. If these structures undergo irreversible or difficult-to-reverse degradation, then even if the human species is not extinct, humanity may already have been deeply weakened as a civilizational subject.

2.3 The Non-Equivalence of Species Survival and Civilizational Subjecthood

The fact that "human beings are still alive" does not mean that "humanity still exists intact as a civilizational subject." There is an important difference between the two. A society may maintain basic population continuity, material production, and institutional shells while gradually losing its ability to judge truth, allocate responsibility, and form autonomous action. Courts may continue to operate, while their factual basis increasingly depends on unverifiable generated content. Schools may continue to teach, while education increasingly trains students to adapt to automated systems rather than to cultivate judgment. Governments may continue to issue policies, while the boundary between human judgment and machine-generated analysis in policy formation becomes increasingly blurred. Firms may continue to make decisions, while the evidence, risk ordering, and comparison of options have already been pre-structured by systems. Individuals may continue to choose, while the space of choice, the structure of reasons, and the definition of problems have already been deeply mediated by technical systems.

In such cases, society has not stopped functioning. It may even appear more efficient, more convenient, more fluid, and more automated. Yet the damage to civilizational subjecthood often does not take the form of external collapse. It appears instead as the transfer, weakening, and hollowing-out of basic capacities. Human beings still act, but increasingly less often form the reasons for action. Human beings still confirm, but increasingly less often complete judgment. Institutions still preserve procedures, but find it increasingly difficult to trace responsibility. Knowledge is still produced, but the relation between truth and verifiability gradually loosens.

Such risks cannot be reduced to "social change" or "cultural discomfort." They concern the basic capacities of the civilizational subject. A civilization continues not only because its members are alive, but because it maintains a common world that can be transmitted, revised, held responsible, verified, and reflected upon. If these capacities are irreversibly weakened, then human existential risk has already moved beyond the species level and entered the level of civilizational subjecthood.

2.4 The Boundary of an Expanded Concept of Existential Risk

To expand existential risk from the species layer to the layer of civilizational subjecthood does not mean that all social problems should be called existential risks. Such an expansion must preserve strict thresholds; otherwise, the concept loses its discriminatory power. Not every institutional friction, cultural transformation, occupational displacement, technological adaptation, or psychological unease constitutes human existential risk. A risk enters the scope of what this paper calls risk to civilizational subjecthood only when it acts on the basic conditions of human civilization and may generate irreversible or difficult-to-reverse structural harm.

For a risk to count as an existential risk at the level of civilizational subjecthood, it must satisfy at least four conditions.

First, it must act on foundational conditions of civilization rather than on local functions. The replacement of a profession, the restructuring of an industry, or the automation of an institutional process does not by itself constitute existential risk. Only when such changes systematically affect human capacities to know truth, allocate responsibility, form judgment, sustain institutions, or create meaning can they enter the scope of existential risk.

Second, it must have the capacity to diffuse across domains. A local error or isolated event is usually insufficient to constitute existential risk. Only when a mechanism can cross education, law, science, media, business, public governance, and everyday life, and repeatedly alter structures of judgment, responsibility, or meaning across multiple systems, does it acquire the features of a risk to civilizational subjecthood.

Third, it must be capable of producing irreversible or difficult-to-reverse structural harm. Existential risk concerns not only scale, but also reversibility. If a mode of technological embedding reshapes education, cognitive habits, institutional procedures, and capacities for agency over the long term, such that several generations gradually lose certain basic capacities, then even without physical destruction it may constitute a deep existential risk.

Fourth, it must affect humanity's capacity to remain a subject of its own future. Existential risk is not any major loss; it is a risk that affects whether humanity can continue to act as the subject of its own future. If a form of AI embedding makes human beings increasingly unable to know truth, govern together, form autonomous judgment, bear responsibility, and create meaning, then it is not merely a problem of social adaptation. It is a problem in which the conditions of civilizational subjecthood are being damaged.

Risk to civilizational subjecthood, then, does not elevate all social difficulties into existential risks. It expands the object of existential-risk assessment from biological continuation to the basic capacities through which humanity exists as a civilizational subject. This expansion neither denies the central importance of species-level risk nor abandons the high threshold that the concept of existential risk requires.

2.5 Summary

A species-centered understanding of existential risk is necessary, but not sufficient. It is necessary because human civilization depends on the human species as its biological substrate. Without the human species, there is no human civilization. It is insufficient because human beings are not only a biological species; they also exist as a civilizational subject. Through language, institutions, science, law, education, culture, responsibility, agency, and structures of meaning, human beings form an intergenerational common world.

This common world is not an accessory to the species. It is a core condition under which human beings become “human.”

The age of AI makes this point unavoidable. AI may in the future threaten the biological survival of humanity, but it is also already entering the epistemic, institutional, judgment-forming, and meaning-making structures of human civilization. AI-related human existential risk must therefore be understood as extending from species survival to civilizational subjecthood. The next section analyzes the first layer of this structure: the human species as the biological substrate of civilization.

3 | The Human Species: The Biological Substrate of Civilization

If the concept of AI-related human existential risk is to be expanded, another misunderstanding must first be avoided. To emphasize humanity as a civilizational subject is not to neglect the foundational role of humanity as a biological species. Civilization is not an abstract entity existing independently of bodies, nervous systems, ecological conditions, and intergenerational continuation. Any human civilization must be borne by embodied human life. For this reason, the human species remains the first layer that any discussion of human existential risk must address.

This paper understands the human species as the existence of *Homo sapiens* as a biological population. This layer includes human bodily structure, nervous systems, perceptual capacities, cognitive capacities, reproductive capacities, physiological vulnerability, mortality, and dependence on ecological and material environments. It constitutes the biological substrate of human civilization. It is not the whole of civilization, but it is the physical precondition for civilization’s emergence and continuation.

This substrate has a double significance. On the one hand, if the human species is destroyed, civilization loses its bearer. On the other hand, the species does not automatically generate civilization. A species may possess a complex nervous system, sociality, and learning capacities without necessarily generating science, law, institutions, philosophy, historical memory, and public reason. Human civilization is possible only with the human species as its bearer; but the human species becomes a historical subject in the specifically human sense only through civilizational structures.

Within the framework of this paper, then, the human species is neither a lower layer that can be bypassed nor a layer sufficient to exhaust human existence. It is necessary but not sufficient.

3.1 Biological Substrate and Embodiment

Human beings, as a species, are first embodied beings. They are not pure information patterns, abstract rational subjects, or arbitrarily transferable computational structures. Human cognition, emotion, judgment, learning, relationships, and responsibility are all closely connected to bodily conditions. Perceiving the world depends on the body. Learning language depends on embodied social experience. Bearing responsibility depends on the capacity to act and on vulnerability to consequences. The formation of meaning is also bound up with birth, death, time, suffering, relations, and finitude.

This means that human civilization is not simply an information system. Civilization contains information, but it is not identical with information. It contains knowledge, but it is not identical with a knowledge base.

It contains institutions, but it is not identical with institutional texts. It contains technologies, but it is not identical with a collection of tools. Civilization requires beings who can learn, judge, remember, commit, act, suffer injury, die, and transmit what they have received. Without embodied human life, civilization would lose its experiential basis, its responsibility structure, and its intergenerational continuity.

Any account that reduces human beings entirely to information patterns therefore fails to explain adequately the substrate of human civilization. Humanity as a civilizational subject still depends on humanity as an embodied species. Species-level risk does not lose its importance when we introduce analysis at the level of civilizational subjecthood. On the contrary, civilizational subjecthood is possible only because it is always borne by embodied human life.

This point must be preserved especially in the context of AI. As AI systems increasingly perform cognitive tasks, generate texts, organize knowledge, and participate in decision-making, human beings may easily be understood as cognitive entities that could be replaced by more efficient information-processing systems. Yet human existence is not merely information processing. Truth, responsibility, agency, and meaning within human civilization are all connected to embodied finitude. A being that dies, can be harmed, makes commitments, regrets, grows over time, and bears consequences does not occupy the same structure of existence as an information system without body, mortality, or vulnerability.

This does not mean that AI cannot participate in human civilization. Nor does it mean that intelligence must exist in a human embodied form in order to have value. The claim of this paper is more limited: if we are to understand the “humanity” that bears the risk, we cannot bypass its biological and embodied basis.

3.2 Species Continuity and Civilizational Continuity

The continuity of the human species is maintained primarily through biological reproduction, intergenerational continuation, and population stability. So long as human beings can reproduce, survive, and maintain basic biological structure, continuity at the species layer remains. Civilizational continuity is different. Civilization depends on the transmission of language, memory, education, institutions, knowledge, norms, technology, value, and meaning. It requires not only that a next generation be born, but also that this generation enter a common world that can be understood, inherited, criticized, and renewed.

Species continuity and civilizational continuity are therefore not automatically equivalent. A society may continue biologically while undergoing civilizational rupture. Throughout history, different civilizational forms have experienced breaks in knowledge, institutional collapse, language loss, technological loss, cultural destruction, and the collapse of structures of meaning. Even if the human species as a whole continues, particular civilizational traditions may disappear. At a deeper level, if such rupture occurs in the basic interfaces on which modern global civilization depends—such as truth-determination, institutional responsibility, subject-forming judgment, and intergenerational transmission of meaning—then it is no longer merely the disappearance of a local tradition. It may become structural damage to human civilization as a whole.

This point means that the discussion of existential risk cannot focus only on population numbers and biological survival. A biologically surviving human world may still be a civilizationally degraded world. People may continue to be born, work, consume, and entertain themselves, while the common world into which they enter can no longer sustain reliable knowledge, responsibility, and the transmission of meaning. If such degradation is irreversible or difficult to reverse, it should be included in a broader analysis of human existential risk.

Under AI conditions, this distinction becomes even more important. AI systems need not threaten human reproduction or biological life systems directly in order to threaten humanity. They are more likely first to alter the mechanisms of civilizational continuity: how knowledge is generated and verified, how education cultivates judgment, how institutions form decisions, how public discussion distinguishes fact from narrative, and how individuals understand the meaning of learning, work, and creation. If these mechanisms are systematically reorganized, the human species may continue while civilizational continuity undergoes a qualitative transformation.

3.3 The Reality of Species-Level Risk

To emphasize risk at the level of civilizational subjecthood is not to imply that species-level risk is secondary. On the contrary, AI may in the future enter species-level risk in multiple ways. If sufficiently powerful AI systems can influence biotechnology, critical infrastructure, autonomous weapons, cyber systems, energy systems, supply chains, or other high-consequence technological systems, they may directly or indirectly threaten the biological survival of humanity. Such risks remain human existential risks in the strictest sense.

The key point is that the reality of species-level risk and the independent importance of risk to civilizational subjecthood are not in conflict. They concern different layers of the object at risk. The former asks whether humanity continues to exist as a biological species. The latter asks whether humanity continues to exist as a civilizational subject. The two may affect one another, but neither can replace the other.

Moreover, civilizational capacities often shape humanity's ability to deal with species-level risk. A civilization that lacks the capacity for truthful judgment cannot reliably evaluate high-consequence technological risks. A civilization with unclear responsibility structures cannot attribute and govern dangerous actions. A civilization with degraded agency cannot form sustained collective action. A civilization whose structures of meaning are shaken may lose the motivation to protect future generations. The protection of biological survival itself therefore depends on capacities at the civilizational and kernel layers.

This shows that risk to civilizational subjecthood does not weaken the importance of species-level risk. It supplements the conditions under which such risk can be understood and managed. If the cognitive, institutional, responsibility-bearing, and agentic capacities of human civilization are weakened, humanity will also be less capable of avoiding genuine biological catastrophe. Species-level risk and risk to civilizational subjecthood should therefore be understood as nested and mutually conditioning, rather than as competing categories.

3.4 The Biological Species Is Not the Whole Human

The definition of humanity as a species cannot by itself answer what gives human life its full significance. Biology can describe human bodies, nervous systems, evolutionary history, and reproductive structures, but it cannot by itself explain legal responsibility, scientific truth, moral commitment, political legitimacy, artistic meaning, or the purposes of education. These belong to the layers of civilization and the philosophical kernel.

A person becomes fully human in the relevant sense not merely by possessing a certain genetic or bodily structure. One must also enter a world of language, relations, norms, education, memory, and meaning. Similarly, a population constitutes a human civilization not merely because its members belong to *Homo sapiens*, but because they jointly sustain a civilizational world that can be transmitted, interpreted, reflected

upon, and renewed.

This point is crucial for AI risk. If an AI system threatens biological bodies, we can describe the threat in the language of traditional species-level risk. If, however, AI systems affect the conditions of language, judgment, responsibility, truth, and meaning, then species-level language is insufficient. We need a higher-level concept through which to describe the risk to humanity as a civilizational subject.

The conclusion of this section can therefore be stated as follows: the human species is the biological substrate of human civilization, but it is not the whole ontology of human existence. To understand AI-related human existential risk adequately, we must analyze the next layer: human civilization.

4 | Human Civilization: The Intergenerational Common World of the Species

If the human species is the biological substrate of human existence, then human civilization is the intergenerational common world formed by that species in history.[Note 4] The “common world” here is not an abstract metaphor. It refers to the shared environment through which human individuals enter structures of language, normativity, knowledge, institutions, responsibility, and meaning. Without this common world, human individuals may still survive as biological organisms, but they cannot become historical subjects, institutional subjects, epistemic subjects, or subjects of meaning in the full sense.

Human civilization is not the simple aggregate of human individuals. Nor is it synonymous with state, culture, technology, or economic system. Civilization is a stable yet changeable structure accumulated by the human species over long historical time. It enables individuals to inherit the languages, memories, sciences, laws, ethics, arts, technologies, educational forms, and narratives of meaning of previous generations, and to continue acting, judging, creating, and taking responsibility within these structures. Civilization is therefore a common environment that extends across individuals, groups, and generations. It does not merely preserve the past; it also organizes the present and makes the future something that can be imagined, committed to, and shaped.

This point is foundational for AI-related human existential risk. If civilization were merely the sum of a population, then so long as human numbers remained, civilization would face no fundamental risk. If civilization were merely state institutions, then the relevant risks would primarily be state failure or political disorder. If civilization were merely cultural expression, then the relevant risks would primarily be changes in lifestyles and narratives of meaning. If civilization were merely a technological system, then the relevant risks would primarily concern the control and distribution of technological capacities. But human civilization is both deeper and wider than all of these concepts. It is the intergenerational operating environment jointly constituted by these systems, and it is the way in which the human species becomes a civilizational subject.

4.1 Civilization Is Not the Sum of Population

Civilization cannot be reduced to population size. The simultaneous existence of many human individuals does not automatically constitute a civilization. Civilization requires a cross-individual and intergenerational structure through which knowledge, norms, institutions, responsibility, and meaning can be preserved, transmitted, criticized, and renewed. Population provides the bearer of civilization, but population by itself does

not automatically generate civilizational continuity.

A society with a large population may still fail to constitute a complex civilization if it lacks stable language, memory, education, institutions, responsibility structures, and knowledge transmission. Conversely, some civilizational traditions may continue to exist through language, texts, education, ritual order, institutional memory, religion, art, and philosophical tradition even after severe losses in population, territory, or political power. Civilizational continuity and population continuity are therefore not automatically equivalent.

This distinction is especially important in the age of AI. AI's impact on human civilization need not first appear as population decline. It may alter the conditions of civilizational continuity without significantly affecting population numbers. For example, if educational systems increasingly cultivate adaptation to automated systems rather than understanding, judgment, and questioning; if knowledge systems increasingly depend on generated content while sources, evidence, and verification processes are marginalized; or if institutional processes increasingly rely on machine ranking, risk assessment, and text generation while responsible subjects become unclear, then population may remain, but the operating conditions of civilization have changed.

Civilizational risk should therefore not be mistaken for population risk. Civilizational risk concerns the sustainability of the common world, not merely the continuation of the population within that world. The human species may continue to reproduce, but if its structures of knowledge, responsibility, judgment, and meaning cannot be stably transmitted, the continuation of humanity as a civilizational subject is damaged.

4.2 Civilization Is Not the State

Civilization also cannot be equated with the state. The state is a political and legal form of organization, typically involving territory, sovereignty, administrative structure, legal system, public finance, and coercive power. Civilization is a deeper, wider, and longer-duration common world. A civilizational tradition may extend across multiple states, and a single state may contain multiple civilizational traditions. Modern states are also embedded in global scientific systems, financial systems, international law, technical standards, digital platforms, and transnational supply chains, none of which belongs entirely within a single state.

Modern science, for example, does not belong to any one state. Modern university systems, medical knowledge, engineering standards, the language of the rule of law, market mechanisms, international trade, financial infrastructure, and internet protocols all cross state boundaries in different ways. States can organize, regulate, protect, or distort these systems, but the state itself is not identical with civilization as a whole. Similarly, Chinese civilization, Indian civilization, Islamic civilization, Western modern civilization, and other civilizational traditions cannot be simply compressed into any single contemporary state.

This distinction is especially important for AI risk. AI risk is often absorbed into the language of national security, industrial competition, regulation, or geopolitics. These forms of language are of course important, because states remain among the most important governing agents in the modern world. But if AI risk is understood only at the level of the state, the deeper impact of AI on the common environment of human civilization will be obscured. AI affects not only the competitive capacity or regulatory power of particular states, but also transnational knowledge production, the global information ecology, digital platforms, public reason, scientific communities, institutional trust, and structures of human meaning.

The state can therefore be understood as a specific deployment of civilization under particular historical, geographical, legal, and political conditions. Civilization is the more fundamental, more durable, and more

complex common world. AI's impact on state governance is an important dimension of AI's impact on civilization, but it is not the whole of it.

4.3 Civilization Is Not Synonymous with Culture

Civilization should also not be reduced to culture. Culture is an important component of civilization, especially in the form of language, customs, art, aesthetics, rituals, stories, identity, habits, and ways of life. Culture gives the deeper values, memories, and meanings of a civilization a form that can be experienced, expressed, and transmitted in everyday life. Without culture, civilization would lose its livable interface.

But civilization includes more than culture. It also includes science, law, institutions, education, economy, technology, public reason, medicine, engineering, administration, infrastructure, and responsibility structures. To understand civilization merely as culture is to underestimate the structures that sustain truth, responsibility, institutions, and knowledge. A civilization may possess rich culture while remaining deeply fragile in science, law, institutions, or responsibility structures. Conversely, a highly technological and institutionalized society may suffer a profound emptiness at the level of culture and meaning.

AI's impact on civilization cannot be reduced to cultural change. AI certainly alters cultural production, including text, images, music, video, games, social expression, and identity narratives. But it also enters knowledge generation, fact verification, legal responsibility, administrative processes, organizational decision-making, educational goals, labor division, and the production of meaning. AI's impact on civilization is therefore not merely a change in cultural style. It is a reorganization of the basic interfaces of the human common world.

More precisely, culture can be understood as the meaning-expression layer and everyday-life interface of civilization. But civilization also includes deeper structures of knowledge, institutions, responsibility, and technology. This distinction is necessary in order to avoid mistaking AI's impact on the conditions of truth, responsibility, agency, and meaning for mere cultural anxiety.

4.4 Civilization Is Not a Technological System

Civilization also cannot be reduced to a technological system. Technology is an important component of civilization, and it often profoundly changes civilizational forms. From agriculture, metallurgy, navigation, printing, electricity, and computing to artificial intelligence, technology continuously expands human capacities for action and organization. Yet civilization is not merely a collection of tools, nor is it a system of efficiency. Civilization also contains value, normativity, responsibility, public reason, meaning, and self-understanding.

Modern technological societies easily equate civilizational progress with capability growth, efficiency improvement, information expansion, and increasing automation. But capability growth does not automatically mean civilizational health. A system may become more efficient while responsibility becomes harder to allocate. It may generate more content while truth becomes harder to verify. It may automate more processes while cultivating human judgment less. It may provide more entertainment and convenience while becoming less able to answer why human beings should continue to create, learn, and bear responsibility.

The age of AI makes it especially important to avoid technological reductionism. AI's significance does not lie merely in the fact that it extends one kind of external tool capacity. It lies in the fact that AI enters multiple

structures at once: knowledge, action, institutions, agency, and meaning. If civilization is understood as the sum of technological capacities, then AI risk will be misunderstood as a single question of whether capability can be controlled. But if civilization is understood as an intergenerational common world, AI risk also concerns whether truth, responsibility, agency, and meaning within that common world continue to hold.

Civilization, then, cannot be defined as the sum of technological capacities. Technology is a powerful system within civilization, but technology by itself does not provide final purposes, define responsibility, guarantee truth, or automatically generate meaning. Without deeper civilizational and philosophical structures, technological growth may produce capability expansion without producing the maturity of the civilizational subject.

4.5 Civilization as an Intergenerational Common World

More precisely, human civilization is an intergenerational common world. Through language, memory, institutions, and education, civilization allows the past to be transmitted into the future and allows finite individual lives to enter a historical structure larger than themselves. A person is not born from nothing. One enters an already existing language, receives inherited institutions and knowledge, and inherits historical memories, scientific achievements, legal structures, ethical norms, and narratives of meaning that one did not personally create. Civilization is this intergenerational structure: it allows human beings to judge, create, and take responsibility on the basis of what previous generations have made possible, and it allows the actions of one generation to matter for future generations.

This intergenerationality gives civilization a vulnerability different from that of the species. Species continuation primarily depends on biological reproduction. Civilizational continuation depends on transmission, understanding, and renewal. If the next generation is born biologically but does not enter a common world capable of cultivating judgment, responsibility, truth, and meaning, then civilizational continuity is weakened. The true continuation of civilization is not merely that “there are people alive,” but that “there are people able to understand, inherit, criticize, and renew a common world.”

AI’s impact on intergenerationality does not lie in preventing the next generation from being born. It lies in changing the path through which the next generation enters the common world. If knowledge acquisition, writing training, problem formulation, and preliminary judgment are increasingly mediated by generative systems, then the relation between the next generation and tradition, texts, evidence, and responsibility will also change. They may still acquire large amounts of content, and may even receive more efficient learning support. But the way they enter the common world has changed: understanding may be replaced by answers; verification by summaries; judgment training by result evaluation; and the exploration, misreading, failure, and insight that belong to tradition may be compressed into immediately generated explanations. This change is not necessarily negative, but it shows that AI affects the mode of civilizational transmission, not merely the tools of learning.

Understanding civilization as an intergenerational common world therefore helps us identify AI risks that do not take the form of physical disaster, yet may alter humanity’s capacity to remain a subject of its future. The continuation of civilization requires more than data preservation, content generation, and institutional operation. It requires human beings who can still enter, understand, judge, and renew the common world.

4.6 Civilization as a Structure of Responsibility

Civilization is not only a system for transmitting knowledge and culture. It is also a structure of responsibility. Through civilization, human beings define actors, commitments, obligations, fault, rights, punishment, compensation, and institutional responsibility. Without a structure of responsibility, complex societies cannot maintain trust, nor can they deal with harm, error, conflict, and commitment.

A responsibility structure depends on many conditions. Actions must be attributable; subjects must be identifiable; norms must be interpretable; evidence must be examinable; procedures must be traceable. If these conditions do not hold, responsibility drifts. Responsibility drift does not mean that there are no consequences. It means that consequences can no longer be stably attributed to an agent, institution, or process.

Once AI enters institutional and action processes, civilization's responsibility structure faces new pressures. When a model generates a recommendation, a platform provides a tool, an organization adopts the output, a human signs off, and an institution executes the result, responsibility no longer easily falls on a single subject. If civilization cannot maintain or reconstruct responsibility structures, it may preserve formal accountability while losing the ability to trace how action is actually formed.

This shows that civilization exceeds population, state, culture, and technology because it also organizes responsibility. A society without a stable responsibility structure may still have technology, culture, and economic activity, but it will be unable to sustain itself as a mature civilizational subject.

4.7 Civilization as an Epistemic Structure

Civilization also provides an epistemic structure. Human beings do not face the world alone and reinvent knowledge independently. They know the world through language, education, scientific method, rules of evidence, documentary systems, peer review, recording practices, and public discussion. Civilization enables individuals to share, criticize, preserve, and revise knowledge.

The core of this epistemic structure is not merely the quantity of information, but verifiability. A civilization can accumulate knowledge because it has mechanisms for distinguishing the more reliable from the less reliable, facts from hypotheses, evidence from rumor, explanation from fabrication. If these mechanisms fail, knowledge production does not necessarily stop. On the contrary, there may be more content and more fluent expression, while the relation between knowledge and truth becomes unstable.

AI is entering this epistemic structure. It can help organize knowledge, but it can also generate content that appears to be knowledge. It can lower the cost of accessing information, but it may also lower the motivation to verify. It can allow more people to express themselves, but it may also fill the content ecology with generated material. Civilization's epistemic structure is therefore facing new conditions of mediation. AI's impact on epistemic structure is not merely a change in information tools. It is a question of how the human common world maintains a shared grasp of truth.

4.8 Civilization as a Structure of Meaning

Finally, civilization is also a structure of meaning. Human beings do not merely want to live. They also need to understand why they live, why they learn, why they create, why they take responsibility, and why

they protect the future. Civilization organizes meaning through religion, philosophy, art, education, family, tradition, political ideals, and public narratives.

Meaning is not an optional psychological decoration. It affects whether individuals are willing to bear responsibility, and whether societies are willing to protect truth, sustain institutions, and act toward the future. A civilization without a structure of meaning may still have material production and technological efficiency, but it will lack the deeper motivations that sustain subjecthood.

AI's impact on structures of meaning is gradually becoming visible. When AI can generate art, text, companionship, research, design, and decision-support, human beings are forced to ask again what creation is, what labor is, what learning is, what contribution is, and what uniqueness is. If these questions cannot be reinterpreted, human beings may fall into a deeper emptiness of meaning even under conditions of greater material abundance and systemic convenience. This shows that the structure of meaning is not a psychological add-on to civilization. It is the structure of reasons through which the civilizational subject continues to act.

4.9 Summary

This section has defined human civilization as the intergenerational common world formed by the human species. Civilization is not the sum of population, not identical with the state, and not merely culture or technology. It includes knowledge, institutions, responsibility, public reason, education, art, economy, law, historical memory, and structures of meaning. It allows the human species to unfold from a mere biological population into a civilizational subject.

This definition grounds the subsequent argument. If human civilization is the common operating environment of the species, then AI's risk to humanity cannot be assessed only by asking whether it threatens human bodies or reproductive capacities. If AI enters structures of knowledge, institutions, responsibility, agency, and meaning, then it may also threaten the conditions under which humanity exists as a civilizational subject.

The next section analyzes why philosophy is not merely one ordinary component of civilization, but the kernel through which civilization defines and revises its foundational categories.

5 | The Philosophical Kernel: The Categorical Structure of Civilizational Self-Understanding and Self-Correction

The previous section understood human civilization as the intergenerational common world formed by the human species. Civilization enables human beings not merely to continue as a biological population, but to exist as a collective subject capable of knowledge, judgment, responsibility, transmission, and meaning-making. Yet civilization itself is not a system that operates without presuppositions. Civilizational systems such as science, law, politics, education, economy, culture, religion, and art can operate only because they presuppose a more basic set of categories: what truth is, what counts as evidence, who is a subject, what responsibility is, what has value, what is worth pursuing, and why human beings should continue to exist.

This paper calls this set of foundational categories, together with the mechanisms through which they are reflected upon, clarified, and revised, the **philosophical kernel** of civilization. "Philosophy" here does not

primarily mean philosophy as an academic discipline, nor does it refer only to theoretical texts written by professional philosophers. It refers to the reflective mechanism through which a civilization understands itself, defines its basic objects, and revises its normative structure.[Note 5] Even a civilization without philosophy departments in the modern academic sense necessarily contains some basic understanding of truth, subjecthood, responsibility, value, and meaning. This understanding may take the form of religion, ritual order, myth, cosmology, ethical tradition, political theory, scientific rationality, public jurisprudence, or practical wisdom. But insofar as a civilization distinguishes truth from falsehood, allocates responsibility, organizes action, evaluates value, and interprets meaning, it is already calling upon something like a philosophical kernel.

Philosophy, in this sense, is not one ordinary application system within civilization. Science, law, economy, politics, education, culture, and art can be understood as functional systems of civilization. Each deals with particular objects, forms particular practices, and possesses its own institutionalized rules. Philosophy deals with the foundational categories that these systems must presuppose in order to operate. Science requires concepts of truth and evidence. Law requires concepts of subject, intention, and responsibility. Politics requires concepts of legitimacy, power, and publicity. Economy requires concepts of value, preference, and the legitimacy of exchange. Education requires concepts of the human being, growth, and the purpose of knowledge. Culture and art require concepts of meaning, expression, and shared experience. If these foundational categories become confused, the functional systems of civilization may still continue to operate, but their outputs will gradually lose a stable basis for interpretation and self-correction.

In this sense, to call philosophy the “kernel” of civilization is not a rhetorical elevation, but a functional claim about levels. “Kernel” does not mean that philosophy is historically prior to civilization in every case, nor does it mean that civilization is deduced from a prewritten philosophical theory. It means that civilizational systems inevitably depend, in their operation, on certain foundational categories concerning existence, truth, responsibility, agency, value, and meaning. When these categories fail, civilization does not merely lose one discipline or one cultural tradition. It loses the basic interface through which it understands, governs, and renews itself.

5.1 Philosophy as the Basis of the Category of Truth

Every civilization must possess some category of truth. By a category of truth, I mean the basic structure through which a civilization distinguishes truth from falsehood, evidence from rumor, knowledge from opinion, explanation from fabrication, and the verifiable from the unverifiable. Without such a structure, civilization cannot stably accumulate knowledge, nor can it sustain science, law, education, historical memory, or public discussion.

The scientific system obviously depends on a category of truth. Science does not merely produce explanations or propose models. It must also be able to say what counts as evidence, what counts as observation, what is reproducible, what is a hypothesis, what is a refutation, and what counts as causal explanation. Without these epistemological presuppositions, scientific activity would degenerate into symbolic production or technical operation, rather than remain a public practice oriented toward truth.

The legal system also depends on a category of truth. Legal judgment must distinguish facts from claims, evidence from narrative, records from fabrication, testimony from speculation. A legal system may possess full procedures, court buildings, judges, and the formal structure of judgment, yet if its factual basis is corrupted and its evidential structure cannot be maintained, the procedural form of law may continue while

substantive justice gradually loses its epistemological foundation.

The educational system likewise depends on a category of truth. Education is not merely the transmission of information. It is the process by which the next generation enters a world of knowledge that can be understood, examined, and criticized. If education becomes the passive consumption of content, students may acquire fluent explanations rather than the capacity to judge truth. In such a case, the form of knowledge remains, but epistemic capacity may be weakened.

Truth, therefore, is one of civilization's foundational categories. Philosophy's role here is to clarify the relations among truth, evidence, knowledge, logic, causality, explanation, and verification. It serves not only science, but the entire civilization's shared grasp of reality. Truth is not an internal function of one application system. It is a foundational category called upon by civilizational systems as such.

5.2 Philosophy as the Basis of the Category of Responsibility

Civilization must not only judge truth; it must also allocate responsibility. Responsibility is not a natural fact. It is a civilizational structure composed of action, subject, intention, consequence, norm, and institution. After an event occurs, civilization must be able to ask: Who acted? Who authorized the action? Who knew? Who should have foreseen the consequence? Who could have prevented it? Who bears the result? Who should be punished or compensated?

These are not merely legal-technical questions. They are philosophical questions. Law can institutionalize responsibility, but law itself depends on a more basic concept of responsibility. Without concepts of agent, intention, negligence, causal attribution, blameworthiness, and obligation, legal procedure cannot operate in a stable way.

In simple cases, responsibility is relatively easy to allocate. A person performs an action, produces a consequence, and an institution assigns responsibility accordingly. Modern complex systems, however, have already made responsibility allocation difficult. Organizations, platforms, algorithms, outsourced processes, expert systems, and institutional procedures jointly participate in producing outcomes, which makes responsibility increasingly distributed. AI further intensifies this difficulty, because AI systems can participate in recommendation generation, evidence selection, option comparison, risk assessment, and action triggering, while lacking the same moral subject status as human beings.

Responsibility is therefore not something that can be fully resolved by legal texts or technical logs alone. Legal texts can assign responsibility. Technical logs can record processes. But questions such as what counts as action, what counts as authorization, who can be attributed with responsibility, and what degree of understanding is sufficient for responsibility still depend on deeper presuppositions in the philosophy of action, legal philosophy, and ethics. If a civilization cannot articulate these presuppositions, even highly complex institutional procedures may preserve form while losing the substance of responsibility.

Philosophy as the basis of the category of responsibility therefore enables civilization to connect action, consequence, and subject. Without this connection, civilization cannot stably deal with error, harm, commitment, obligation, and accountability.

5.3 Philosophy as the Basis of the Category of Agency

Civilization must also define what it means to be a subject or agent. A subject is not simply a biological individual, nor is it any system capable of producing outputs. Subjecthood involves the capacity to form judgment, bear responsibility, participate in action, occupy a standpoint, and enter normative relations. Different civilizations define subjecthood differently, but every civilization must possess some category of agency or subjecthood. Without it, law, politics, education, ethics, and responsibility cannot operate.

The category of agency answers a set of basic questions: Who can judge? Who can consent? Who can refuse? Who can authorize? Who can bear responsibility? Who can represent themselves? Who is merely a tool? Who is the true source of action? In modern civilization, these questions appear in concepts such as personhood, rights, responsibility-capacity, free will, legal subjecthood, political subjecthood, and moral agency.

The age of AI makes the category of agency especially tense because AI systems increasingly participate in the processes that precede human judgment. They help human beings define problems, organize materials, compare options, generate reasons, simulate consequences, and propose recommendations. In such cases, human beings may still appear to be choosing, but the conditions of choice, the structure of reasons, and the visible set of options may already have been mediated by systems.

This does not mean that all AI assistance weakens agency. The use of tools is itself part of human civilization. The problem arises when a tool not only extends action but begins to pre-organize judgment. Civilization must then distinguish assistance from replacement, recommendation from judgment, use from dependence, confirmation from agency. Philosophy as the basis of the category of agency helps civilization make these distinctions.

Agency is not a naturally given object that requires no definition. Human beings can be understood within civilization as responsible subjects, acting subjects, and subjects of meaning only because civilization already possesses a categorical structure of agency. AI's mediation of the judgment-forming process requires that this structure be examined anew.

5.4 Philosophy as the Basis of the Category of Value

Civilization does not only deal with facts and responsibility. It must also order multiple values. Any complex society simultaneously pursues safety, freedom, efficiency, truth, dignity, justice, innovation, stability, prosperity, and meaning. These values often conflict. A civilization must decide which value takes priority under which circumstances, which values may not be sacrificed, which goals are merely instrumental, and which goals have intrinsic importance.

Such value ordering cannot be completed by science or technology alone. Science can provide knowledge about consequences. Technology can provide means for achieving goals. Economy can measure costs and incentives. But none of them can by itself determine what is worth pursuing. The question of what should be prioritized is itself a question of value theory and ethics.

AI systems introduce new pressure at this point. AI often operates in the form of optimization. Given goals, metrics, reward functions, user feedback, or organizational performance indicators, AI can greatly improve efficiency. But optimization by itself does not show whether the goal is justified, nor whether the metric being optimized truly represents human value. A system can optimize engagement while weakening public

reason. It can optimize productivity while damaging agency. It can optimize convenience while reducing human judgment. It can optimize satisfaction while concealing poverty of meaning.

Civilization therefore requires a category of value through which it can judge what technological capacities should serve. If civilization lacks the capacity to order values, it easily mistakes measurability for importance, efficiency for the good, convenience for progress, and user preference for human value. Philosophy does not replace science or technology. It gives civilization the capacity to judge the legitimacy of its ends.

5.5 Philosophy as the Basis of the Category of Meaning

Civilization must also answer questions of meaning. Meaning is not an accessory to individual psychology. It is the structure through which individual action, responsibility, sacrifice, learning, creation, and commitment to the future become intelligible. Human beings do not seek only survival and efficiency. They also ask why one should learn, why one should work, why one should create, why one should bear responsibility, why one should protect the future, why one should seek truth, and why human beings still matter.

These questions cannot be answered by technological capacity itself. Even if AI helps human beings complete more work, generate more content, provide more entertainment, and optimize more processes, it cannot automatically answer why human beings should continue to bear responsibility and create meaning. On the contrary, precisely because AI may approximate or replace an increasing number of human capacities, questions of meaning become more acute.

If labor ceases to be the source of value for most people, if education is no longer primarily the transmission of knowledge, if creation no longer expresses a distinctively human capacity, and if judgment is increasingly assisted or pre-organized by systems, then human beings must reinterpret the basis of their own value. This cannot be fully resolved by employment policy, nor can it be absorbed by mental health services alone. It belongs to the reconstruction of civilization's structure of meaning.

Philosophy as the basis of the category of meaning enables civilization to ask again, under changing conditions of capacity, what grounds human value. Does human value depend on capability? Does dignity depend on efficiency? Must creation be irreplaceable in order to have meaning? Can responsibility be outsourced? Is the purpose of education merely to acquire usable knowledge? If these questions are not asked, civilization may become materially more convenient while becoming poorer in meaning.

Meaning is therefore one of civilization's foundational categories, not a mere byproduct of individual subjective feeling. If civilization cannot sustain the category of meaning, it weakens the structure of reasons through which individuals bear responsibility, pursue truth, participate in institutions, and act toward the future.

5.6 Philosophy as Civilization's Mechanism of Self-Correction

Civilization does not only need to operate. It also needs to correct itself. Every civilization makes mistakes. Scientific theories are overturned. Legal systems lag behind reality. Political orders become corrupt. Cultural traditions become rigid. Economic systems lose balance. Technological trajectories produce unforeseen side effects. If civilization is to continue, it must possess some capacity to reflect upon and revise itself.

Philosophy does not provide a fixed set of answers here. It provides civilization with the capacity to reflect on

its own presuppositions. It allows civilization to ask: How do we know that we know? Are our institutions legitimate? Are our concepts of responsibility adequate to new structures of action? Has our ordering of values been captured by instrumental reason? Is our understanding of the human too narrow? Are we mistaking generation for truth, formal choice for agency, capability growth for civilizational progress?

This reflective capacity is different from local error correction. Local correction can be performed through technical optimization, policy adjustment, or institutional reform. Philosophical reflection addresses deeper presuppositions: what counts as error, what counts as evidence, who has authority to judge, why responsibility is needed, what is worth protecting. Deep civilizational crisis often consists not in application-layer error, but in the destabilization of these foundational categories.

The age of AI is precisely such a situation. AI does not merely introduce new tools. It also changes how civilization judges tools, subjects, responsibility, and meaning. If civilization cannot reflect on these foundational categories, it may interpret new realities through old concepts and thereby fail both to diagnose risks accurately and to form effective responses.

Philosophy is therefore the kernel mechanism through which civilization understands, adjusts, and renews itself. Without such a mechanism, civilization may continue to operate, but it will depend increasingly on inertia, efficiency, and automation rather than understanding and reflection. Such a civilization may become faster, but not necessarily more lucid.

5.7 The Relation Between the Philosophical Kernel and AI Existential Risk

Understanding philosophy as civilization's kernel helps explain why AI-related human existential risk cannot be understood only at the level of application or tool use. If AI changes only the efficiency of a tool, it is primarily a technical risk. If AI affects only the workflow of a particular industry, it is primarily an industrial risk. If AI produces only certain problems of content quality, it is primarily an information-governance risk. But if AI begins to change how civilization defines truth, responsibility, agency, value, and meaning, then it touches the kernel layer of civilization.

This does not mean that every AI application constitutes a kernel-layer risk. A concept loses explanatory power when overgeneralized. The key question is whether AI's embedding destabilizes civilization's foundational categories, or makes human beings increasingly unable to maintain the reflective capacity through which these categories are revised. At that point, the object of risk is not merely whether a system performs well. It is whether civilization can still understand itself, correct itself, and sustain the foundational categories that allow humanity to continue as a civilizational subject.

If human civilization is supported by foundational categories such as truth, responsibility, agency, value, and meaning, then systematic AI-driven disturbances to these categories cannot be classified simply as application-layer problems. Before discussing technical performance, product safety, institutional governance, or social impact, we must first ask whether these changes are touching the conditions under which humanity can continue to exist as a civilizational subject. In this sense, the philosophical kernel is not an external interpretive layer. It is a conceptual foundation that AI risk analysis itself must confront.

5.8 Summary

This section has argued for the significance of philosophy as civilization's kernel. Philosophy here is not a narrow academic discipline, but the reflective structure through which civilization defines and revises its foundational categories. Civilizational systems such as science, law, politics, economy, education, culture, and art all depend on categories such as truth, responsibility, agency, value, and meaning. Philosophy is the mechanism through which these categories are made explicit, clarified, and corrected.

Understanding philosophy as the kernel of civilization helps explain why AI risk cannot be understood only at the level of application or tool use. If AI merely changes the efficiency of tools, it is primarily a technical risk. If AI enters categories such as truth, responsibility, agency, value, and meaning, then it may become a kernel-layer risk. If humanity exists as a civilizational subject, then the foundational categories that sustain civilizational subjecthood become unavoidable objects of existential-risk analysis.

The next section uses the functional analogy of hardware, kernel, and distribution to clarify further the layered relation among the human species, the philosophical kernel, human civilization, and the functional systems of civilization.

6 | Hardware-Kernel-Distribution: A Functional Analogy and Its Limits

The preceding sections have distinguished three layers: the human species, human civilization, and the philosophical kernel. The human species is the biological substrate on which civilization becomes possible. Human civilization is the intergenerational common operating environment formed by the species. Philosophy is the reflective kernel through which civilization defines foundational categories such as truth, responsibility, agency, value, and meaning. To clarify the relation among these three layers, this section introduces a functional analogy: hardware, kernel, and distribution.

The analogy must be qualified from the beginning. It is not literal. Human beings are not machines, civilization is not software, and philosophy is not code. The analogy is not intended to technologize human civilization or replace philosophical analysis with engineering language. Its purpose is to clarify functional relations among different layers: some layers provide physical bearing, some provide foundational categorical and rule-like interfaces, and some constitute complete operating environments that can be historically deployed, updated, inherited, and localized.

In computing systems, the term “operating system” is often imprecise. It may refer to the kernel, to the complete user environment, or to a particular distribution. In the present analogy, the kernel refers to the underlying foundational interface, while the distribution refers to the complete user environment organized around that kernel. To avoid conceptual confusion, this paper uses a more fine-grained analogy: the human species is like hardware; the philosophical kernel is like the Linux kernel; and human civilization is closer to a distribution such as Ubuntu, Debian, or Red Hat Enterprise Linux—that is, a complete operating environment containing the kernel, system tools, libraries, permission structures, software ecosystems, user interfaces, default configurations, and practical habits.[Note 6]

The value of this analogy lies in its ability to prevent two confusions. First, the existence of hardware does not mean that the system is operating well. Second, the fact that a distribution can boot does not mean

that the kernel is healthy. Similarly, the continued existence of human bodies does not mean that humanity remains intact as a civilizational subject. The continued operation of social systems does not mean that their foundational categories remain stable. The production of outputs by functional systems such as science, law, economy, education, and culture does not mean that foundational interfaces such as truth, responsibility, agency, value, and meaning remain healthy.

6.1 The Human Species as Hardware

Within the three-layer structure of human existence, the human species corresponds to the physical substrate layer. In the computer-system analogy, it is like hardware. Hardware provides the material basis on which a computing system can operate. Without processor, memory, storage, power supply, bus, and input-output devices, no kernel, distribution, or application can actually run. Similarly, without human bodies, brains, nervous systems, perceptual capacities, reproductive mechanisms, and dependence on material environments, human civilization cannot occur.

This analogy helps clarify the foundational role of the species layer. Human civilization is not an informational cloud existing independently of human bodies. It depends on embodied human life. Language learning, emotional development, educational transmission, ethical relations, responsibility-bearing, and meaning formation are all connected to human bodily conditions, temporality, finitude, and vulnerability. The human species as biological substrate is therefore not an object that can be replaced by analysis at the civilizational layer.

Yet hardware is not the whole system. A computer without a kernel, distribution, file system, user environment, and applications is not a usable computing environment. Similarly, the human species by itself does not automatically constitute human civilization. The biological body provides the bearer, but civilization also requires language, institutions, knowledge, norms, memory, and structures of meaning. The species layer is a necessary condition, but not a sufficient condition.

This point is especially important for analyzing AI-related human existential risk. If AI directly threatens human bodies, ecological conditions, or reproductive capacities, then it threatens the species layer. Such risks are certainly existential risks in the most basic sense. But if AI does not directly damage human bodies and instead alters how civilization judges truth, allocates responsibility, forms agency, and understands meaning, then it threatens a higher-level system structure. The hardware may remain, yet the system may no longer operate well.

6.2 Philosophy as the Linux Kernel

If the human species can be compared to hardware, philosophy is more appropriately compared to a kernel. In a Linux system, the Linux kernel is not the complete system that users ordinarily operate directly, but it is the foundational layer through which the system runs. It manages hardware resources, provides system-call interfaces, maintains permission boundaries, coordinates processes and resources, and enables upper-level tools and applications to operate in stable ways. Users usually do not interact with the kernel directly. They enter a complete operating environment through distributions such as Ubuntu, Debian, or Red Hat Enterprise Linux. Yet if the kernel layer develops serious problems, the stability of the entire system is affected.

In civilization, philosophy as kernel does not mean that some fixed philosophical doctrine governs all civiliza-

tions, nor that academic philosophy stands institutionally above science, law, politics, or culture. It means that any civilization, in order to operate, must implicitly contain a set of foundational categories through which it defines truth, evidence, responsibility, agency, value, and meaning. These categories cannot be completely determined by application-layer systems, because those systems must already call upon them in order to exist.

Science must call upon interfaces of truth and evidence. Law must call upon interfaces of subjecthood and responsibility. Politics must call upon interfaces of legitimacy and publicity. Economy must call upon interfaces of value and preference. Education must call upon interfaces concerning the human being, knowledge, and growth. Culture must call upon interfaces of meaning and expression. If these underlying interfaces become confused, the functional systems of civilization may continue to operate, but their outputs will become increasingly difficult to interpret, attribute, or correct.

From this perspective, philosophy can be called the kernel of civilization because it deals with civilization's deepest categorical structure. A civilization may lack an explicit philosophical discipline, but it cannot lack implicit determinations of truth, responsibility, agency, value, and meaning. Philosophy is the mechanism through which these determinations are reflected upon, made explicit, criticized, and revised.

6.3 Human Civilization as a Distribution-like Operating Environment

If philosophy is the kernel, then human civilization should not simply be called an “operating system” without further distinction. In ordinary language, “operating system” sometimes means the kernel, sometimes a distribution, and sometimes the whole user environment. To avoid conceptual collapse, this paper compares human civilization more specifically to a distribution or complete operating environment.

In the Linux ecosystem, Ubuntu, Debian, and Red Hat Enterprise Linux are not merely the Linux kernel. They include system tools, libraries, package managers, file-system organization, default permission settings, user interfaces, network configuration, drivers, documentation, application ecosystems, and habits of use. Different distributions may share certain kernel mechanisms while presenting different user experiences, governance structures, software ecosystems, and default value orientations.

Human civilization is similar. Civilization is not merely the philosophical kernel, nor merely science, law, or technological applications. It is a complete operating environment that includes language, institutions, law, science, education, economy, technology, art, religion, public reason, historical memory, narratives of meaning, and practical traditions. It translates foundational philosophical categories into a common world that can be lived, institutionalized, educated into, transmitted, contested, and revised.

This analogy also helps make sense of civilizational diversity. Different civilizational traditions may share certain foundational categorical problems—truth, responsibility, agency, value, and meaning—while organizing them through different institutions, languages, ritual orders, knowledge systems, and cultural forms. Similarly, different Linux distributions may share the Linux kernel while forming different system styles, default configurations, and modes of use. Differences among civilizations cannot be reduced simply to differences at the kernel layer, but neither can we ignore the common pressure placed on foundational categories.

Human civilization as a distribution or complete operating environment is therefore the historical unfolding between the human species and the philosophical kernel. It depends on the human species as hardware,

calls upon the foundational categorical interfaces supplied by the philosophical kernel, and forms a livable common world through institutions, knowledge, culture, technology, and education.

6.4 Human Civilization as a Whole: A Shared Ecology of Multiple Civilizational Traditions

The scale of the expression “human civilization” requires further clarification. Human civilization, as used in this paper, is not identical with any particular state, culture, or historical civilizational tradition. Nor does it universalize one concrete form of civilization as humanity as such. Human civilization should be understood as an overall civilizational ecology composed of multiple civilizational traditions. Different traditions may have different languages, religions, institutions, legal orders, knowledge traditions, artistic forms, and narratives of meaning. Yet together they constitute the intergenerational common world formed by the human species in history.

Within the engineering analogy, particular civilizational traditions may be understood as different distributions or system configurations. They may share certain foundational categorical problems—such as truth, responsibility, agency, value, and meaning—while answering them in different ways. Chinese civilization, Indian civilization, Islamic civilization, Western modern civilization, and other traditions are not simple copies of one and the same distribution. They are civilizational operating environments formed under different historical conditions, each with its own institutional configurations, knowledge structures, cultural interfaces, and systems of meaning.

But human civilization as a whole is not the mechanical sum of these traditions. It is closer to a civilizational ecology composed of multiple distributions, shared protocols, cross-dependencies, historical memories, knowledge exchanges, experiences of conflict, and institutional interfaces. Modern science, international law, global trade, financial networks, universities, public-health systems, internet protocols, digital platforms, and AI infrastructure have made different civilizational traditions interdependent to an unprecedented degree. Human civilization as a whole therefore includes both difference and common structure, both plurality and certain global operating conditions.

This is crucial for AI-related human existential risk. AI’s impact on civilization does not operate only within one civilizational tradition. It may also act on the common interfaces of human civilization as a whole: how knowledge is produced and verified, how institutions allocate responsibility, how technology enters governance, how public discussion maintains a shared reality, and how human beings understand their own value and future. Once these interfaces are systematically disturbed, the impact may cross states, cultures, and regions, becoming a risk at the level of human civilization as a whole.

The civilizational subject referred to in this paper is therefore not a single cultural community, a state subject, or a single civilizational tradition. It is humanity as an overall civilizational subject extending across generations, institutions, knowledge systems, and structures of meaning. It includes the differences among multiple civilizational traditions, as well as the human-civilizational whole constituted by them. AI-related human existential risk requires us to understand the bearer of risk at this overall scale.

6.5 Civilizational Functional Systems as the Application Layer

Once human civilization as a whole is understood as an ecology of multiple civilizational traditions, we must also clarify the role of its functional systems. Science, law, politics, economy, education, art, religion, media, and technology can be understood as functional systems or application layers within civilizational distributions. Each performs particular tasks: science produces knowledge; law allocates responsibility; politics organizes power; economy coordinates resources; education transmits capacities; art expresses meaning; religion sustains ultimate concern; media organizes public attention; technology extends capacities for action.

These functional systems are not secondary. On the contrary, civilization operates concretely through them. Ordinary people usually do not encounter the “philosophical kernel” directly. They enter civilization through schools, courts, laboratories, markets, media, families, religions, art, firms, and governments. Similarly, users usually do not interact directly with the Linux kernel. They use the interfaces, tools, permission settings, and application ecosystems provided by distributions such as Ubuntu, Debian, or Red Hat Enterprise Linux.

Within particular civilizational traditions, these functional systems take different configurations. Different civilizations may have different legal traditions, educational goals, religious structures, artistic forms, political organizations, and narratives of meaning. Like tools, default configurations, and user habits preinstalled in different distributions, these configurations express their respective historical paths and value orderings. Civilizational functional systems therefore do not float abstractly above humanity. They are configured within concrete histories, languages, institutions, and cultures.

Under modern conditions, however, many functional systems have also formed cross-civilizational and global interfaces. Modern scientific communities, international law, global financial networks, universities, public-health systems, internet protocols, supply chains, digital platforms, and AI infrastructure do not belong entirely to one state or civilizational tradition. They flow, connect, conflict, and merge across civilizational traditions, becoming a functional layer of human civilization as a whole. Disturbances to these systems may therefore no longer remain local; they may diffuse through global interfaces into human civilization as a whole.

Application-layer systems, however, cannot operate independently of foundational categories. If science loses its interface to truth and evidence, it degenerates into symbolic production. If law loses its interface to subjecthood and responsibility, it preserves procedure while losing accountability. If education loses its categories of the human being and growth, it becomes the distribution of skills or adaptation to systems. If economy loses value judgment, it mistakes measurability for importance. If art loses structures of meaning, it becomes content production. If technology loses boundaries of purpose, it mistakes capability growth for civilizational progress.

The health of civilizational functional systems therefore depends not only on their own efficiency, but also on whether they can still stably call upon foundational categories. Application-layer flourishing does not equal kernel-layer health. There may be more content while knowledge becomes less reliable; faster processes while responsibility becomes less clear; more choices while agency becomes thinner; richer entertainment while meaning becomes poorer.

In this sense, the flourishing of the application layer does not automatically prove the health of the civilizational system. Content can increase, processes can accelerate, interfaces can become smoother, and user satisfaction can rise; none of this by itself shows that foundational interfaces such as truth, responsibility, agency, value, and meaning remain stable. The age of AI requires special attention to a situation in which civilizational functional systems become more efficient while the pressures on their foundational categories

become harder to perceive. The smoother a system becomes, the more users may trust that it is healthy; the more convenient civilization becomes, the more difficult it may be for human beings to notice whether they still retain the capacities for judgment, responsibility, and meaning-formation.

AI's impact on civilizational functional systems must be understood at this scale. AI is not entering only a product, an industry, or a national institution. It is entering scientific writing, legal documents, educational training, corporate decision-making, media dissemination, artistic creation, administrative processes, and public discussion. It may improve the efficiency of these systems, but it may also change how they call upon foundational categories such as truth, responsibility, agency, value, and meaning. AI's impact is therefore not merely an application-layer change. It may exert pressure on the civilizational kernel through the application layer.

6.6 AI's Different Pressures on the Three-Layer Structure

In the above analogy, AI does not enter only one application context, nor does it affect only a local system within one civilizational tradition. Through functional systems such as science, law, education, economy, media, administration, business, and everyday life, AI is entering the operating environment of human civilization as a whole. AI's pressure on human existence must therefore be understood in layers: it may act on the species layer, the civilizational layer, and the kernel layer. Risks at these different layers cannot replace one another, nor should they be conflated.

When AI enters biotechnology, critical infrastructure, autonomous weapons, cyber systems, or other high-consequence chains of real-world action, it may act on the species layer by affecting the biological survival of humanity and the openness of its future. In this case, the risk is directed at the hardware layer: whether humanity continues to exist as an embodied species.

When AI enters education, law, scientific research, administration, business, media, public services, and everyday life, it acts on the civilizational layer. It may change how knowledge is organized, how institutions form judgments, how responsibility is allocated, how public reason is sustained, and how culture and meaning are produced. In this case, the risk is not direct bodily destruction, but the reorganization of the operating environment of the civilizational distribution.

When AI changes the stability of foundational categories such as truth, evidence, responsibility, agency, value, and meaning, it acts on the kernel layer. At this level, the issue is not merely whether one institutional process has failed, or whether one model output is reliable. The issue is whether civilization can still understand itself, correct itself, and maintain the conceptual interfaces through which humanity continues to exist as a civilizational subject.

This layered distinction explains why AI-related human existential risk cannot be understood only as risk of species extinction, nor only as a problem of product safety or regulatory compliance. AI may not first destroy the hardware, but may alter the distribution. It may not immediately destroy the distribution, but may contaminate or disturb kernel interfaces. If kernel-layer categories fail, application-layer systems may still continue to operate, but they will increasingly struggle to interpret their own outputs, attribute their own responsibilities, and understand their own purposes.

6.7 The Limits of the Analogy

Every analogy has limits. The hardware–kernel–distribution analogy is not meant to reduce human civilization to a computer system, nor to suggest that civilization has a single, closed, fully designable kernel. Human civilization is more open, more historical, more conflictual, and more unpredictable than a computer system. The philosophical kernel is not a central control module, but a set of foundational categories that are continually disputed, interpreted, revised, and institutionalized through history.

Moreover, a computer kernel is usually maintained by a limited engineering team, whereas the philosophical kernel of civilization has no single maintainer. It is distributed across religion, law, philosophy, science, education, politics, art, and public discussion. It is updated through contestation rather than compilation, becomes effective through practice rather than command, and continues through intergenerational transmission rather than version release. The analogy should therefore not be pushed to the literal level.

What the analogy aims to express is only three functional relations. First, civilization needs biological bearing, just as a system needs hardware. Second, civilizational functional systems need foundational categorical interfaces, just as applications need kernel services. Third, complete civilizational life is not the kernel itself, but a historically formed operating environment composed of kernel, tools, institutions, practices, and user environments—just as a distribution such as Ubuntu is not the Linux kernel itself, but a complete system-world organized around that kernel.

Within these limited terms, the analogy is useful. It helps us see that human existential risk may occur not only at the hardware layer, but also at the distribution layer and the kernel layer. The human species may continue to exist while the civilizational distribution becomes unhealthy. Civilizational functional systems may continue to operate while the philosophical kernel becomes unstable. The age of AI forces us to attend to all three layers at once.

6.8 Summary

This section has used the functional analogy of hardware, kernel, and distribution to clarify the relation among the human species, the philosophical kernel, and human civilization. The human species is like hardware: the biological bearer of civilization. The philosophical kernel is like the Linux kernel: the foundational interface of civilizational categories. Particular civilizational traditions can be understood as different distributions or system configurations. Human civilization as a whole is an overall civilizational ecology constituted by multiple civilizational traditions, shared protocols, cross-civilizational functional systems, and global interfaces.

The purpose of the analogy is not to technologize humanity, but to clarify differences among layers. Hardware can remain intact while the system fails to operate well. A distribution can boot while the kernel is damaged. Applications can flourish while foundational interfaces become unstable. Similarly, the human species may continue to exist while humanity is no longer intact as a civilizational subject. Institutions and technical systems may continue to operate while foundational categories such as truth, responsibility, agency, value, and meaning become unhealthy.

The next section explains why the development of AI makes this three-layer structure not merely a philosophical analysis, but a problem that AI-related human existential risk theory must confront.

7 | Why AI Makes This Three-Layer Structure a Problem of Existential Risk

The previous sections distinguished three layers—the human species, human civilization, and the philosophical kernel—and explained their relation through the functional analogy of hardware, kernel, and distribution. This structure is not intended to construct an abstract anthropological schema. Its purpose is to clarify what AI-era human existential risk may act upon. AI makes this three-layer structure important because it does not merely affect a tool, industry, or application domain. It is entering the connections among the human species, the civilizational operating environment, and the foundational categorical structure of civilization.

If AI were merely an external tool, its risks could be analyzed primarily through product safety, usage norms, industry regulation, or engineering reliability. But current AI systems increasingly mediate how human beings acquire knowledge, form judgment, organize action, execute institutions, and understand themselves. AI is not merely adding another tool. It is entering the operating environment of human civilization. It does not affect only an application layer; it may also alter how civilizational functional systems call upon foundational categories such as truth, responsibility, agency, value, and meaning. AI risk therefore cannot be understood at only one layer.

7.1 AI and the Species Layer

At the species layer, AI-related risk is most easily captured by the traditional language of existential risk. If sufficiently powerful AI systems can influence biotechnology, critical infrastructure, autonomous weapons, cyber systems, energy systems, supply chains, or other high-consequence chains of real-world action, they may directly or indirectly threaten humanity's biological survival and the openness of its future. Such risks concern whether humanity continues to exist as an embodied species, and therefore belong to human existential risk in the most direct sense.

This point must be preserved. To emphasize risk at the level of civilizational subjecthood does not mean that species-level risk becomes secondary. On the contrary, the human species is the biological bearer of civilization. If the species layer is destroyed, civilization cannot continue. Any expanded risk framework must recognize the foundational status of species extinction, permanent disempowerment, and the foreclosure of the future.

Yet AI's impact on humanity does not necessarily first take this form. Many AI systems at the current stage do not directly act on human bodies or reproductive capacities. Instead, they first enter informational, judgment-forming, institutional, and action environments. By changing how human beings know, organize, authorize, and act, they indirectly affect humanity's capacity to manage high-consequence technological risks. Species-level risk is therefore foundational, but it does not exhaust the object of human risk in the age of AI.

7.2 AI and the Civilizational Layer

At the civilizational layer, AI's impact appears as a reorganization of the common operating environment of humanity. AI systems are entering civilizational functional systems such as education, scientific research, law, medicine, administration, business, media, art, public services, and everyday life. They can help orga-

nize information, generate texts, compare options, select evidence, assist judgment, automate processes, and coordinate action. These functions have real value, and may greatly expand human capacities.

But precisely because AI is useful, it will be widely absorbed into civilizational systems. If a tool remains merely an external addition, its impact is usually limited to local use contexts. But when a tool becomes an interface to knowledge, an institutional process, a mediator of organizational judgment, and a condition of everyday action, it begins to alter the operating environment of civilization itself. At this point, the risk is no longer merely whether a particular system output is correct. It is whether structures of knowledge, institutions, responsibility, and judgment in the human common world are being reorganized.

At this layer, AI may allow civilization to continue functioning on the surface while gradually altering its internal conditions. Schools still teach, courts still adjudicate, firms still decide, governments still issue policies, media still produce content, and individuals still make choices. Yet the definition of problems, the organization of materials, the selection of evidence, the generation of reasons, the ordering of risks, and the available paths of action may increasingly be pre-structured by AI systems. Civilization has not stopped operating, but its mode of operation may be undergoing deep change.

The key issue here is not merely the degree of automation, but the position of human beings in the process by which action is formed. If human beings gradually retreat from the full chain of defining problems, gathering evidence, comparing reasons, and forming judgment to the terminal position of confirming, approving, or consuming results, then civilizational subjecthood has already undergone a structural shift. Such a shift does not require malicious design. It may be driven naturally by efficiency, convenience, and user-experience optimization. It does not appear as human exclusion from the process. It appears as human beings remaining in the process while participating less and less in the front-end formation of judgment.

This shows that AI's impact on the civilizational layer cannot be reduced to efficiency improvement or industrial transformation. Efficiency improvement by itself does not constitute existential risk. The problem arises when efficiency improvement is accompanied by structural changes in how knowledge is verified, responsibility is allocated, judgment is formed, and meaning is transmitted. At that point, AI begins to touch the conditions under which humanity can continue to exist as a civilizational subject. Human civilization is not merely a task-completion system. It is a common world that enables human beings to understand, judge, take responsibility, transmit, and create meaning.

7.3 AI and the Kernel Layer

At the kernel layer, AI's impact is deeper still. The issue is not merely whether some institutional process has adopted AI, or whether some class of tasks has been automated. The issue is whether the foundational categories through which civilization defines truth, responsibility, agency, value, and meaning are being placed under pressure.

When AI-generated content increasingly resembles knowledge, civilization must again distinguish generation, explanation, evidence, and truth. A system's ability to provide fluent explanations does not mean that it provides verifiable knowledge. If human beings increasingly take generated content as the starting point of knowing, rather than sources, evidence, and processes of verification, then the category of truth is placed under pressure.

When AI participates in recommendation generation, evidence selection, risk assessment, and action triggering, civilization must again understand responsibility and authorization. The fact that a human subject

signs the final document does not necessarily mean that the subject understands the process by which the judgment was formed. The fact that an institution retains approval and audit procedures does not necessarily mean that it can trace how action was formed. The category of responsibility therefore faces new structural conditions.

When AI pre-organizes problems, options, reasons, and paths of action for human beings, civilization must again understand agency. Agency cannot be equated simply with a final click, confirmation, or choice. If human beings retain the form of choice but increasingly participate less in problem definition, evidential judgment, and reason formation, then the category of agency requires renewed clarification.

When AI can generate works, research, designs, companionship, and decision-support, civilization must also understand value and meaning anew. If capability, efficiency, and output no longer stably distinguish human beings from machines, then human beings must again ask what grounds creation, learning, labor, responsibility, and dignity. The category of meaning is no longer merely a question of individual psychology; it becomes part of civilization's self-understanding.

These changes show that AI may touch the kernel layer of civilization. It does not merely change outputs at the application layer. It may also change how civilizational functional systems call upon foundational categories. Once these categories become unstable, civilization may continue to maintain institutional and technical operation, while becoming increasingly unable to explain its own knowledge, responsibility, agency, and purposes.

7.4 The Interdependence of the Three Layers

The species layer, the civilizational layer, and the kernel layer are not three isolated objects. They are nested and interdependent. The species layer provides biological bearing. The civilizational layer provides an intergenerational common operating environment. The kernel layer provides the foundational categories through which civilization understands, revises, and renews itself. Deep damage at any one layer may affect the other layers.

Species-level risk terminates the biological bearer of civilization. This is the most direct existential harm. Civilizational-layer risk weakens humanity's shared capacity to organize knowledge, institutions, responsibility, and meaning, thereby affecting the capacity for collective action. Kernel-layer risk weakens civilization's capacity to understand itself, correct itself, and define its own ends, so that even if civilization continues to operate, it may become unable to judge whether it remains healthy.

This interdependence is especially important under AI conditions. A civilization lacking the capacity for truthful judgment will struggle to identify high-consequence technological risks. A civilization with unclear responsibility structures will struggle to govern complex chains of action. A civilization with weakened agency will struggle to sustain autonomous judgment over time. A civilization whose structures of meaning are shaken may lack reasons to protect the future. Protecting the human species itself therefore depends on capacities at the civilizational and kernel layers. Protecting civilizational subjecthood, in turn, cannot be separated from the biological bearing provided by the species layer.

AI-related human existential risk therefore cannot be compressed into a single-layer problem. It may simultaneously concern species survival, civilizational continuity, and the stability of foundational categories. A risk framework that focuses only on the species layer will underestimate damage to the conditions of civilizational subjecthood. A framework that focuses only on social adaptation will underestimate species-level

risk and kernel-layer categorical pressure. A complete analysis must see all three layers at once.

7.5 From the Three-Layer Structure to the Bearer of Risk

This three-layer structure ultimately returns us to the central question of this paper: What is the bearer of AI-related human existential risk? If the bearer of risk is only *Homo sapiens* as a biological species, then risk analysis will primarily ask whether humanity becomes extinct, whether it is permanently disempowered, and whether it loses its biological future. This layer is indispensable, but it is not sufficient.

If the bearer of risk also includes humanity as a civilizational subject, then risk analysis must also ask whether human beings can still maintain a common world that is transmissible, verifiable, accountable, reflective, revisable, and capable of sustaining meaning. Human beings are not merely living organisms. They are historical subjects living within structures of language, institutions, responsibility, knowledge, and meaning. If AI changes these structures, it may alter the conditions under which humanity can continue to exist as a civilizational subject.

This paper therefore does not expand AI risk indefinitely into all forms of social change. It clarifies the layered structure of the bearer of risk. Species-layer risk concerns whether humanity continues to exist. Civilizational-layer risk concerns whether humanity can continue to exist as the subject of a common world. Kernel-layer risk concerns whether human civilization can continue to understand, revise, and renew its foundational categories. Together, these three layers constitute the concept of the bearer of risk required for the analysis of AI-related human existential risk.

7.6 Summary

This section has explained why the three-layer structure of human species, human civilization, and philosophical kernel becomes a problem that existential risk theory must confront in the age of AI. AI may act on the species layer, affecting humanity's biological survival and the openness of its future. It may act on the civilizational layer, reorganizing structures of knowledge, institutions, responsibility, judgment, and meaning. It may also act on the kernel layer, placing foundational categories such as truth, responsibility, agency, value, and meaning under pressure.

This does not mean that every AI application constitutes an existential risk, nor that all social change should be included within the concept of existential risk. The key point is that when AI's impact touches the basic conditions under which humanity can continue to exist as a civilizational subject, and may produce irreversible or difficult-to-reverse structural harm, risk analysis can no longer remain only at the species layer or the application layer. The next section proposes an expanded definition of AI-related human existential risk on this basis.

8 | Redefining AI-Related Human Existential Risk: From Species Survival to the Civilizational Subject

The preceding argument can be brought together as a task of conceptual engineering: to redefine the bearer of risk in AI-related human existential risk. In conceptual engineering, the task is not only to analyze how existing concepts are used, but also to revise concepts under new theoretical and practical pressures so that they better serve explanation, judgment, and action (Chalmers, 2025; Haslanger, 2005). If “humanity” is understood only as *Homo sapiens* as a biological species, then AI-related human existential risk primarily concerns whether humanity will be physically destroyed, permanently disempowered, deprived of its biological future, or in some decisive sense lose control over its own future. This understanding captures the most direct and foundational forms of existential risk, but it does not exhaust the object of risk in the age of AI.

Humanity also exists as a civilizational subject. Through language, knowledge, institutions, responsibility, truth, agency, value, and meaning, human beings form an intergenerational common world. This common world is not an ornament added to the species. It is a condition under which human beings exist as historical subjects, epistemic subjects, responsible subjects, and subjects of meaning. If AI affects only external tools or local industries, it need not constitute human existential risk. But if AI irreversibly weakens the conditions under which humanity can continue to exist as a civilizational subject, then even if the human species survives, the risk may have existential significance.

This paper therefore proposes the following expanded definition:

AI-related human existential risk refers to risks arising from the development, deployment, embedding, or loss of control of AI systems that may irreversibly destroy the human species, permanently deprive humanity of its future, or irreversibly weaken the basic conditions under which humanity can continue to exist as a civilizational subject.

This definition preserves the central importance of species-level existential risk while expanding the bearer of risk from the biological species alone to the civilizational subject. It does not require that all AI-driven social change be included within existential risk. Nor does it require that every disturbance to civilization be treated as existential harm. It claims only that when AI’s impact touches the basic conditions under which humanity continues as a civilizational subject, and when this impact has irreversible or difficult-to-reverse structural consequences, existential risk theory cannot assess the risk solely in terms of biological survival.

8.1 Why an Expanded Definition Is Needed

The need for an expanded definition arises from AI’s distinctive position. Many technologies alter human capacities for action, but they do not directly alter civilization’s foundational categories. AI is distinctive because it is entering structures of knowledge, institutions, judgment, action, and meaning. It may not only perform tasks, but also mediate how human beings define problems, acquire evidence, form reasons, allocate responsibility, and understand value and meaning.

Under these conditions, a traditional species-centered definition encounters a blind spot. Suppose AI does not directly kill human beings and does not immediately destroy their reproductive capacities, but makes human beings increasingly unable to verify truth, allocate responsibility, form judgment, sustain institutions,

or create meaning. Such harm can easily be downgraded to a social problem, cultural change, difficulty of adaptation, or lag in governance. Yet once these capacities are basic conditions for humanity's continued existence as a civilizational subject, their irreversible weakening cannot be treated as ordinary social change.

The expanded definition is not intended to lower the threshold of existential risk. It is intended to make the bearer of risk more complete. Human existential risk matters not only because humanity may disappear as a biological species, but also because humanity may be irreversibly weakened as a civilizational subject capable of knowing, judging, taking responsibility, creating meaning, and acting toward the future. If this latter form of harm is excluded from existential risk theory, AI risk analysis will systematically underestimate deep risks that do not take the form of bodily extinction.

8.2 Species-Level Risk Remains Foundational

The expanded definition does not weaken the seriousness of species extinction, permanent disempowerment, or the foreclosure of the future. On the contrary, it takes these risks as foundational. If the human species becomes extinct, the civilizational subject cannot continue. If humanity permanently loses the capacity to shape the future, human civilization also loses the historical space in which it can continue to unfold. Species-level risk therefore remains the most direct and irreplaceable form of human existential risk.

The revision proposed by this paper is not that risk to civilizational subjecthood should be placed above species-level risk. Rather, the two belong to different layers. Species-level risk asks whether humanity continues to exist as a biological population. Risk to civilizational subjecthood asks whether humanity continues to exist as a collective subject capable of knowledge, governance, judgment, responsibility, and meaning-making. The former is the biological precondition of the latter. The latter is the civilizational condition under which the former can be understood as a human future rather than mere biological continuation.

The expanded definition therefore does not replace the traditional concept of existential risk. It supplements the traditional definition by identifying a risk object that may be overlooked under AI conditions. It preserves species destruction as the baseline, while also arguing that if the basic conditions of humanity as a civilizational subject may be irreversibly weakened, existential risk theory must include them within its scope of analysis.

8.3 Thresholds for Risk to Civilizational Subjecthood

To avoid conceptual overextension, risk to civilizational subjecthood must have clear thresholds. Not every institutional change, cultural transformation, occupational displacement, technological adaptation, or psychological unease caused by AI constitutes human existential risk. For a risk to enter the scope of existential risk at the level of civilizational subjecthood, it must satisfy at least four conditions.

First, it must act on foundational civilizational conditions rather than on local functions. The restructuring of an industry, the replacement of a profession, or the automation of an institutional process does not automatically constitute existential risk. Such changes acquire civilizational-existential significance only when they systematically affect humanity's capacity to know truth, allocate responsibility, form judgment, sustain institutions, or create meaning.

Second, it must have the capacity to diffuse across domains. Local errors, single accidents, or isolated misuses usually do not constitute existential risks. A mechanism becomes a risk to civilizational subjecthood

only when it can cross education, science, law, media, business, public governance, and everyday life, and repeatedly alter human structures of judgment, responsibility, or meaning across multiple systems.

Third, it must be capable of producing irreversible or difficult-to-reverse structural harm. Existential risk concerns not only the magnitude of loss, but also reversibility. If a mode of technological embedding reshapes education, cognitive habits, institutional procedures, and capacities for agency over the long term, such that several generations gradually lose certain basic capacities, then even without physical destruction it may constitute a deep existential risk.

At the civilizational scale, irreversibility does not always mean absolute impossibility of recovery. It may instead mean that the costs of recovery, intergenerational losses, institutional path dependence, and breakdowns in the training and transmission of capacities reach a level sufficient to alter humanity's capacity to remain the subject of its own future. Once certain capacities cease to be trained and transmitted over a long period, their loss may not appear as deprivation; later generations may no longer experience the loss as a loss. Even if such harm is not metaphysically irreversible, it may become historically extremely difficult to reverse.[Note 7]

Fourth, it must affect humanity's capacity to remain a subject of the future. Existential risk is not any major harm. It is a risk that affects whether humanity can continue to act as the subject of its own future. If a form of AI embedding makes human beings increasingly unable to know truth, govern together, form autonomous judgment, bear responsibility, and create meaning, then it is not merely a problem of social adaptation. It is a problem in which the conditions of civilizational subjecthood are being damaged.

These four conditions preserve the boundary of the expanded definition. They prevent existential risk from being reduced to species extinction alone, while also preventing all social problems from being elevated into existential risk. Risk to civilizational subjecthood has existential significance not because it sounds grave, but because it acts on the basic capacities through which humanity continues to exist as a civilizational subject.

8.4 Why This Is Not an Ordinary Social Risk

Risk to civilizational subjecthood differs from ordinary social risk in its object, structure of diffusion, and reversibility. Ordinary social risks may be serious, but they usually affect a particular institution, industry, group, region, or period. They may produce great losses without necessarily changing the basic conditions under which humanity continues to exist as a civilizational subject.

AI's impact on the conditions of civilizational subjecthood is not necessarily weakening in every case. The literature on AI alignment and meaningful human control has shown that AI systems may also play constructive roles in value clarification, responsibility tracking, and the expansion of human control capacities (Gabriel, 2020; Santoni de Sio & van den Hoven, 2018). Risk arises in another kind of situation: when such mediation replaces rather than enhances human foundational capacities over the long term, and when it makes the training, transmission, and recovery of these capacities difficult.[Note 8]

The key feature of risk to civilizational subjecthood is that it acts on the basic capacities of the civilizational common world: the capacity to know truth, allocate responsibility, form judgment, sustain institutions, and create and transmit meaning. These capacities are not the functions of any single department. They are conditions under which humanity can continue to act as a civilizational subject. If they are systematically

weakened, society may still operate, technology may still advance, and population may still continue, but humanity's capacity to remain the subject of its own future will be weakened.

This kind of risk is also different from ordinary cultural anxiety. Cultural anxiety often arises from discomfort with change, and does not necessarily have existential significance. The assessment of risk to civilizational subjecthood cannot be based on nostalgia, stylistic preference, or instinctive resistance to technological change. It must be based on structural conditions: whether a change weakens human capacities to know, judge, take responsibility, and create meaning; whether such weakening diffuses across domains; whether it is difficult to reverse; and whether it affects humanity's capacity to act as a future-bearing subject.

Risk to civilizational subjecthood is therefore not a moralization of social change, nor a pathologization of technological adaptation. It is a stricter question: whether AI may alter the basic conditions under which humanity can continue to exist as a civilizational subject.

8.5 Why This Is Still Existential Risk

A possible objection is that, so long as humanity is not extinct, such risks should not be called existential risks. On this view, damage to truth, responsibility, agency, or meaning—however serious—should be treated as social, ethical, or cultural risk rather than existential risk.

This objection rightly preserves the high threshold of existential risk, but it narrows the object of “human existence” too much. Human existence is not merely the continuation of biological organisms. Humanity's existence as a civilizational subject includes the capacities to know truth, form institutions, bear responsibility, exercise judgment, create meaning, and act toward the future. If these capacities are irreversibly weakened, the damage does not concern only one institution or cultural field. It concerns humanity's mode of existence as a future-bearing subject.

Existential risk is distinguished from ordinary major risk not only by the scale of harm, but also by whether it affects the openness and subjecthood of the human future. If the human species continues to exist, but its capacities for truth, responsibility, judgment, and meaning are systematically weakened to the point that humanity can no longer act as the subject of its own future, then this condition differs from species extinction, but still has deep existential significance.

Risk to civilizational subjecthood is therefore not an arbitrary expansion of the concept of existential risk. It is a more complete understanding of the object called “human existence.” It requires us to distinguish biological continuation from the continuation of civilizational subjecthood, while recognizing that both are conditions of the human future.

8.6 Implications for AI Risk Theory

This expanded definition has three direct implications for AI risk theory.

First, it changes the bearer of risk. AI risk analysis should not ask only whether AI may destroy the human species. It must also ask whether AI may irreversibly weaken the conditions under which humanity can continue to exist as a civilizational subject. The object of risk expands from a biological population to a nested subject constituted by embodied species, civilizational common world, and philosophical kernel.

Second, it changes how risks are identified. Some AI risks may not first appear as catastrophic physical events. They may appear instead as structural degradation of civilizational capacities. Knowledge may still be produced, institutions may still operate, content may still be generated, and choices may still exist, while the underlying conditions of truth, responsibility, agency, and meaning are already under pressure. Risk analysis must therefore attend to processes of formation, structural conditions, and reversibility, rather than only to final catastrophic outcomes.

Third, it changes the premise of governance. If AI risk is understood only as species-level risk, governance primarily concerns the prevention of catastrophic loss of control, physical destruction, and permanent disempowerment. If AI risk also includes risk to civilizational subjecthood, governance must also ask how truth is maintained, how responsibility is attributed, how agency is preserved, how meaning is transmitted, and how civilization retains the capacity for self-correction. Governance is no longer only the constraint of external technical systems. It also includes the protection of the foundational conditions under which humanity exists as a civilizational subject.

These implications do not constitute a concrete risk classification, nor do they provide a concrete governance program. They show only that once the bearer of risk has been redefined, the scope of analysis, standards of identification, and premises of governance in AI risk theory must change accordingly.

8.7 Summary

This section has proposed an expanded definition of AI-related human existential risk. The definition preserves the foundational importance of traditional species-level risk while also recognizing that humanity exists as a civilizational subject. If AI irreversibly weakens the civilizational conditions under which humanity can continue to exist as a subject of truth, responsibility, agency, and meaning, then even if AI does not directly destroy the human species, it may constitute a deep form of existential risk.

This expanded definition does not classify all AI-driven social change as existential risk. It maintains conceptual boundaries through four thresholds: foundational civilizational conditions, cross-domain diffusion, structural harm, and capacity for future subjecthood. Its central significance is that the starting point of AI-related human existential risk analysis cannot remain at species survival alone. It must also include the conditions under which humanity can continue to exist as a civilizational subject.

9 | Conclusion: From Species Survival to the Civilizational Subject

This paper began from a prior question: when we say that AI may pose an existential risk to humanity, what exactly is the “humanity” that is at risk? This question matters because the definition of the bearer of risk determines the scope of risk analysis. If humanity is understood only as *Homo sapiens* as a biological species, then AI-related human existential risk will primarily be understood as species extinction, permanent disempowerment, biological foreclosure of the future, or decisive loss of control. This understanding captures the most basic and serious forms of existential risk, but it does not exhaust the object of human risk in the age of AI.

This paper has not denied the importance of species-level risk. On the contrary, it has consistently treated the human species as the biological substrate of civilization. Without embodied human life, nervous sys-

tems, perceptual capacities, reproductive mechanisms, mortality, and intergenerational continuation, human civilization cannot continue to unfold. Species survival is therefore a necessary condition of the human future. What this paper adds is that a necessary condition is not a sufficient condition. Human beings may continue to exist biologically while losing, at the civilizational level, certain basic capacities that enable the continuation of a human common world.

To clarify this point, the paper distinguished three nested layers. First, the human species is the biological bearer of civilization. Second, human civilization is the intergenerational common world formed by the human species in history, including language, knowledge, institutions, responsibility, public reason, historical memory, and structures of meaning. Third, the philosophical kernel is the reflective structure through which civilization defines and revises its foundational categories, including truth, evidence, responsibility, agency, value, and meaning. Through this structure, humanity does not merely continue as a biological population. It exists as a civilizational subject capable of knowledge, judgment, responsibility, transmission, and meaning-making.

The paper further used the functional analogy of hardware, kernel, and distribution to clarify this layered structure. The human species is like hardware; the philosophical kernel is like the Linux kernel; and human civilization is like a distribution or complete operating environment such as Ubuntu, Debian, or Red Hat Enterprise Linux. The point of this analogy is not to technologize humanity, but to show that hardware can remain intact while the system fails to operate well; a distribution can still boot while the kernel is damaged; and applications can continue to flourish while foundational interfaces become unstable. Similarly, the continued survival of the human species does not mean that humanity remains intact as a civilizational subject. The continued operation of institutions and technical systems does not mean that foundational categories such as truth, responsibility, agency, value, and meaning remain healthy.

On this basis, the paper proposed an expanded definition: AI-related human existential risk refers to risks arising from the development, deployment, embedding, or loss of control of AI systems that may irreversibly destroy the human species, permanently deprive humanity of its future, or irreversibly weaken the basic conditions under which humanity can continue to exist as a civilizational subject. This definition preserves the central importance of species-level existential risk while expanding the bearer of risk from the biological species alone to the civilizational subject.

This expansion does not mean that all AI-driven social change constitutes existential risk. A risk must act on foundational civilizational conditions, have the capacity to diffuse across domains, be capable of producing irreversible or difficult-to-reverse structural harm, and affect humanity's capacity to remain a subject of its own future in order to enter the scope of existential risk at the level of civilizational subjecthood. Without these thresholds, serious social problems may remain important risks, but they need not be existential risks. Risk to civilizational subjecthood has existential significance not because it sounds grave, but because it acts on the basic capacities through which humanity continues to exist as an epistemic subject, a responsible subject, an acting subject, and a subject of meaning.

This conceptual reconstruction has direct significance for AI risk theory. It requires us, before asking whether AI will destroy humanity, to clarify the layered structure of the "humanity" at stake. It also requires us, before asking whether AI systems are reliable, controllable, or compliant, to ask whether these systems are changing the conditions under which human beings know, judge, take responsibility, and create meaning. AI risk should not be understood only as the possibility that an external technical system causes catastrophe. It should also be understood as the possibility that AI changes the conditions under which humanity can continue to exist as a civilizational subject.

In the context of existential risk, conceptual clarification is not an external or merely preliminary task. It is a condition of risk identification. If human beings cannot name the capacities they are losing, they will struggle to determine which changes should be governed, protected, or constrained. One difficulty in the age of AI is that many deep harms may not appear as disasters. They may enter civilization in the form of efficiency, convenience, automation, and normal operation. Clarifying the bearer of risk is therefore necessary in order to give these less visible forms of harm an ontological position in which they can be discussed, compared, and scrutinized.

The contribution of this paper is therefore not a complete risk taxonomy, nor a governance program. It is a conceptual reconstruction of the bearer of risk. It moves the starting point of AI-related human existential risk analysis from species survival alone to the conditions under which humanity can continue to exist as a civilizational subject. Only on this basis can the specific types, mechanisms, and governance implications of AI risk be given clearer ontological coordinates.

Endnotes

[Note 1] On AI systems participating in AI development itself, research automation, and recursive self-improvement, see Favaro and Clark (2026), published by the Anthropic Institute, as well as related discussions of AI-driven R&D, AI-assisted AI development, and recursive self-improvement. The use of this background does not imply that AGI has already been achieved, nor that recursive self-improvement has already been completed. It is cited here to indicate that AI is increasingly entering its own development chain, thereby giving AI-related human existential risk a new structural background.

[Note 2] On AI in relation to human dignity, personhood, responsibility, the common good, and public ethics, see recent work in AI ethics, AI governance, human dignity, and public-ethical discourse, including UNESCO's *Recommendation on the Ethics of Artificial Intelligence* (2022) and Pope Leo XIV's encyclical on safeguarding the human person in the time of artificial intelligence. These sources are not used to ground the paper in any particular religious or theological premise. Rather, they indicate that AI has entered public-ethical domains concerning human self-understanding, responsibility, and dignity.

[Note 3] On classic discussions of existential risk, see Bostrom (2013) on existential risk and Ord (2020) on existential catastrophe and humanity's long-term future. On contemporary discussions of AI x-risk, superintelligence, the control problem, and alignment, see Bostrom (2014), Russell (2019), Center for AI Safety (2023), Carlsmith (2022), Yudkowsky and Soares (2025), and Ngo, Chan, and Mindermann (2022). Related community discussions include Christiano's (2019) analysis of gradual failure scenarios and Karnofsky's (2021) discussion of the technological significance of the present century. This paper accepts the foundational importance of species extinction, permanent disempowerment, future foreclosure, and loss-of-control risks, while asking whether the "humanity" to which these risks refer should be understood only as a biological species.

[Note 4] The paper's account of civilization as an intergenerational common world can be placed in dialogue with Arendt's (1958) discussion of world, action, and plurality; Cassirer's (1944) analysis of symbolic forms and human culture; Taylor's (2004) theory of social imaginaries; and MacIntyre's (1981) account of tradition and practices. The paper does not identify the concept of the civilizational subject with any one of these theories. It draws on these traditions to clarify that human civilization is not a simple aggregate of population, state, culture, or technology, but a common world through which human beings inherit, judge, take responsibility, and renew themselves.

[Note 5] The “philosophical kernel” does not refer to philosophy as a modern academic discipline. It refers to the reflective mechanism through which a civilization defines, clarifies, and revises its foundational categories. Relevant background includes conceptual engineering (Chalmers, 2025; Haslanger, 2005), philosophy of technology (Verbeek, 2011; Vallor, 2016), philosophical anthropology, social epistemology, and philosophy of information (Floridi, 2011). Different civilizational traditions need not make their foundational categories explicit in the form of “philosophy” as a modern academic field; these categories may also appear through religion, ritual order, cosmology, ethical tradition, public jurisprudence, or practical wisdom.

[Note 6] The hardware–kernel–distribution analogy is a functional analogy, not a literal identity. The examples of the Linux kernel, Ubuntu, Debian, and Red Hat Enterprise Linux are used to help distinguish hardware, kernel, distribution, and application layer. The paper does not claim that human civilization can be reduced to a computer system, nor that philosophy has a single maintainer in the way an engineering kernel might. The analogy only clarifies the layered difference among physical bearing, foundational categorical interfaces, and complete historical operating environments.

[Note 7] “Irreversible or difficult-to-reverse” should not, at the civilizational scale, be understood only as absolute impossibility of recovery. It refers instead to recovery costs, intergenerational losses, institutional path dependence, and breakdowns in the training and transmission of capacities reaching a level that alters humanity’s capacity to remain the subject of its own future. Relevant background includes path dependence, institutional persistence, intergenerational transmission, skill formation, and human capital depreciation. Pierson’s (2000) account of institutional path dependence and Arthur’s (1989) analysis of technological lock-in and increasing returns are especially relevant for understanding the institutional and technological mechanisms of difficult reversibility.

[Note 8] This paper does not assume that AI can only weaken the conditions of civilizational subjecthood. AI may also enhance knowledge organization, error detection, responsibility tracking, public reflection, and institutional auditing. The paper focuses on another possibility: when AI mediation replaces rather than enhances human foundational capacities over the long term, and when it makes the training, transmission, and recovery of those capacities difficult. Relevant discussions include human agency, meaningful human control, human-AI collaboration, AI augmentation, and the responsibility gap. See, for example, Matthias (2004) on the responsibility gap, Santoni de Sio and van den Hoven (2018) on meaningful human control, Gabriel (2020) on AI alignment and values, and Ngo, Chan, and Mindermann (2022) on the alignment problem from a deep learning perspective.

References

- Arendt, H. (1958). *The human condition*. University of Chicago Press.
- Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394), 116–131. <https://doi.org/10.2307/2234208>
- Bostrom, N. (2013). Existential risk prevention as global priority. *Global Policy*, 4(1), 15–31. <https://doi.org/10.1111/1758-5899.12002>
- Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press.
- Carlsmith, J. (2022). *Is power-seeking AI an existential risk?* arXiv. <https://doi.org/10.48550/arXiv.2206.13353>

- Cassirer, E. (1944). *An essay on man: An introduction to a philosophy of human culture*. Yale University Press.
- Center for AI Safety. (2023, May 30). *Statement on AI risk*. <https://www.safe.ai/statement-on-ai-risk>
- Chalmers, D. J. (2025). What is conceptual engineering and what should it be? *Inquiry*, 68(9), 2902–2919. <https://doi.org/10.1080/0020174X.2020.1817141>
- Christiano, P. (2019, March 17). *What failure looks like*. AI Alignment Forum. <https://www.alignmentforum.org/posts/HBxe6wdjxK239zajf/what-failure-looks-like>
- Favaro, M., & Clark, J. (2026). *When AI builds itself: Our progress toward recursive self-improvement, and its implications*. Anthropic Institute. <https://www.anthropic.com/institute/recursive-self-improvement>
- Floridi, L. (2011). *The philosophy of information*. Oxford University Press.
- Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines*, 30(3), 411–437. <https://doi.org/10.1007/s11023-020-09539-2>
- Haslanger, S. (2005). What are we talking about? The semantics and politics of social kinds. *Hypatia*, 20(4), 10–26. <https://doi.org/10.1111/j.1527-2001.2005.tb00533.x>
- Karnofsky, H. (2021). *The “most important century” series*. Cold Takes. <https://www.cold-takes.com/most-important-century/>
- Leo XIV. (2026, May 15). *Magnifica humanitas: On safeguarding the human person in the time of artificial intelligence* [Encyclical letter]. The Holy See. <https://www.vatican.va/content/leo-xiv/en/encyclicals/documents/20260515-magnifica-humanitas.html>
- MacIntyre, A. (1981). *After virtue: A study in moral theory*. University of Notre Dame Press.
- Matthias, A. (2004). The responsibility gap: Ascribing responsibility for the actions of learning automata. *Ethics and Information Technology*, 6(3), 175–183. <https://doi.org/10.1007/s10676-004-3422-1>
- Ngo, R., Chan, L., & Mindermann, S. (2022). *The alignment problem from a deep learning perspective*. arXiv. <https://doi.org/10.48550/arXiv.2209.00626>
- Ord, T. (2020). *The precipice: Existential risk and the future of humanity*. Hachette Books.
- Pierson, P. (2000). Increasing returns, path dependence, and the study of politics. *American Political Science Review*, 94(2), 251–267. <https://doi.org/10.2307/2586011>
- Russell, S. (2019). *Human compatible: Artificial intelligence and the problem of control*. Viking.
- Santoni de Sio, F., & van den Hoven, J. (2018). Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI*, 5, Article 15. <https://doi.org/10.3389/frobt.2018.00015>
- Taylor, C. (2004). *Modern social imaginaries*. Duke University Press.
- UNESCO. (2022). *Recommendation on the ethics of artificial intelligence*. UNESCO. <https://unesdoc.unesco.org/ark:/48223/pf0000381137>

Vallor, S. (2016). *Technology and the virtues: A philosophical guide to a future worth wanting*. Oxford University Press.

Verbeek, P.-P. (2011). *Moralizing technology: Understanding and designing the morality of things*. University of Chicago Press.

Yudkowsky, E., & Soares, N. (2025). *If anyone builds it, everyone dies: Why superhuman AI would kill us all*. Little, Brown and Company. <https://www.hachettebookgroup.com/titles/eliezer-yudkowsky/if-anyone-builds-it-everyone-dies/9780316595643/>

Appendix A: Core Concepts and Terminological Notes

This appendix defines the core concepts used in the paper. Its purpose is not to introduce new arguments, but to stabilize the terminology required for the paper's conceptual framework.

AI-related human existential risk

AI-related human existential risk refers to risks arising from the development, deployment, embedding, or loss of control of AI systems that may irreversibly destroy the human species, permanently deprive humanity of its future, or irreversibly weaken the basic conditions under which humanity can continue to exist as a civilizational subject.

Human existential risk

Human existential risk is the broader category of risk that concerns whether humanity can continue to exist, retain its future, and remain the subject of that future. In this paper, it includes but is not exhausted by species extinction.

Bearer of risk

The bearer of risk is the subject or object that is exposed to existential harm. This paper argues that the bearer of AI-related human existential risk should not be understood only as *Homo sapiens* as a biological species, but also as humanity as a civilizational subject.

Human species

The human species refers to *Homo sapiens* as a biological population. It includes human bodies, nervous systems, reproductive capacities, physiological vulnerability, mortality, and dependence on ecological and material environments.

Human civilization

Human civilization is the intergenerational common world formed by the human species in history. It includes language, knowledge, institutions, law, science, education, economy, culture, technology, responsi-

bility structures, historical memory, and structures of meaning.

Civilizational subject

A civilizational subject is humanity understood not merely as a biological population, but as an intergenerational bearer of knowledge, institutions, responsibility, agency, value, and meaning. It is the subject that can know, judge, take responsibility, transmit, and create meaning across generations.

Civilizational subjecthood

Civilizational subjecthood refers to the condition under which humanity can continue to exist and act as a civilizational subject. It depends on the stability and transmission of truth, responsibility, agency, value, meaning, institutions, and common-world structures.

Risk to civilizational subjecthood

Risk to civilizational subjecthood refers to risks that may weaken, irreversibly or in ways difficult to reverse, the conditions under which humanity can continue as a civilizational subject. It does not include all social change, but only structural damage that crosses strict thresholds.

Intergenerational common world

The intergenerational common world is the shared world into which individuals are born and through which language, memory, knowledge, institutions, responsibility, and meaning are transmitted, criticized, and renewed across generations.

Philosophical kernel

The philosophical kernel is the reflective structure through which civilization defines, criticizes, and revises its foundational categories. It is not identical with academic philosophy as an institution, nor with a fixed doctrine. It is a functional layer of civilizational self-understanding and self-correction.

Foundational categories

Foundational categories are the basic categories presupposed by civilizational systems. In this paper, the most important are truth, evidence, responsibility, agency, value, and meaning.

Species layer

The species layer is the biological layer of human existence. It concerns the survival, reproduction, embodiment, vulnerability, and material continuity of *Homo sapiens*.

Civilizational layer

The civilizational layer is the layer of the intergenerational common world. It concerns the operation of knowledge, institutions, law, education, culture, responsibility, and meaning.

Kernel layer

The kernel layer is the layer of foundational categories. It concerns whether civilization can still define, criticize, and revise categories such as truth, responsibility, agency, value, and meaning.

Hardware–kernel–distribution analogy

The hardware–kernel–distribution analogy is a functional analogy used to clarify the relation among the human species, the philosophical kernel, and human civilization. The human species is like hardware; the philosophical kernel is like a kernel; human civilization is like a distribution or complete operating environment. The analogy is not literal and has no reductionist intent.

Application layer

The application layer refers to civilizational functional systems such as science, law, politics, economy, education, art, religion, media, and technology. These systems perform concrete functions, but they depend on foundational categories.

Civilizational continuity

Civilizational continuity is the continued transmission, understanding, and renewal of the common world across generations. It is not equivalent to biological population continuity.

Civilizational irreversibility

Civilizational irreversibility does not necessarily mean absolute impossibility of recovery. It refers to cases in which recovery costs, intergenerational losses, institutional path dependence, and breakdowns in capacity transmission become so deep that humanity's capacity for future subjecthood is altered.

Path dependence

Path dependence refers to the way historical choices, institutions, infrastructures, and habits can make certain future paths increasingly difficult to reverse. It helps explain why some civilizational harms may become difficult to undo even when they are not physically irreversible.

AI augmentation

AI augmentation refers to cases in which AI strengthens rather than replaces human capacities, such as knowledge organization, error detection, responsibility tracking, public reflection, and institutional auditing.

Meaningful human control

Meaningful human control refers to the condition under which human beings remain meaningfully involved in the control, authorization, and accountability of autonomous or AI-mediated systems. It is relevant to agency, responsibility, and governance.

Responsibility gap

The responsibility gap refers to the difficulty of attributing responsibility for actions or outcomes produced by autonomous or learning systems when no human agent fully controls or understands the process.